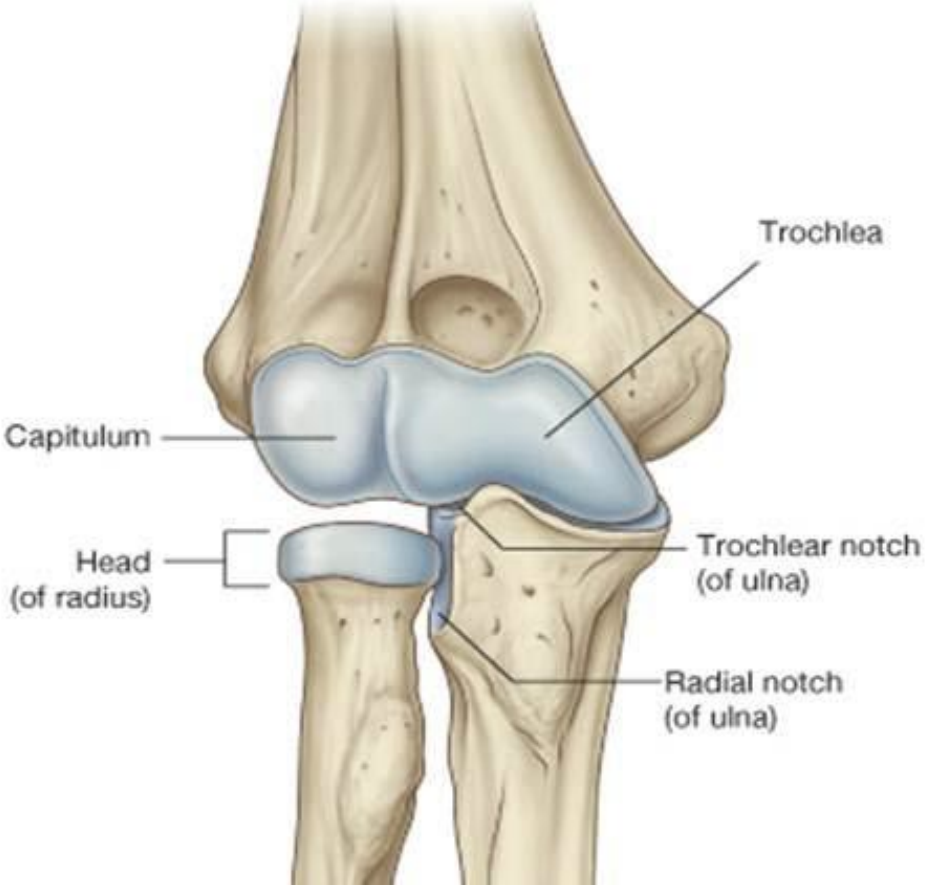
COURSE

Elbow Joint

Proffesor Dr Galal Mansour

10 lectures



Watch course

15 October 2026

TRANS PRO

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IMPORTANT

The world is changing very fast, especially with the rapid development of Artificial Intelligence (AI). We believe we should keep up with these changes and use them to make learning easier and more accessible.

This work was created using a computer program developed by us through Vibe Coding, together with several other AI-assisted tools. Using these tools, we were able to create transcripts of lectures, including lectures delivered in Egyptian Arabic, and organize them into a more academic, book-like format.

The aim is to help the listener focus on listening without having to write notes, while having an organized transcript they can easily return to for quick revision.

This transcript is not a replacement for the original lecture or the relevant textbooks and scientific references. The accuracy is generally very good, but AI-assisted transcription can still contain errors. If you are unsure about any point, please check the original lecture or a reliable reference.

This work is completely non-commercial and was created simply to help those seeking knowledge and make educational resources more accessible.

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How to use this book

- Click any line in the Contents to jump to that lecture.
- The time at the bottom right of each page (» 04:57) opens the original lecture on YouTube at that exact moment.
- "↑ Chapter" returns to the lecture's contents; "↑ Book" returns to the book's contents.
- Click any section heading to return to the lecture's contents.
- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

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Elbow Joint Course: Part 1

(Surgical Anatomy, Articular Congruity and the Fat Pad Sign)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 16) “carrying angle smaller in females” → “carrying angle smaller in males”

Course Overview Ask AI

- Scope: **elbow joint** conditions, traumatic and pathological
- Audience: orthopaedic surgeons
  - Diploma, master's and doctorate (MD) students
- Teaching method: each anatomical point → linked to its clinical relevance

Elbow Joint Composition and Congruity Ask AI

Trochoginglymoid Joint Ask AI

Elbow

- Elbow = **trochoginglymoid joint**
  - Three joints in one capsule
  - Biomechanically different from each other
- Not separate joints → shared capsule and synovium

Joint	Proximal / ulnar side	Distal humeral / radial side
Ulnohumeral	Greater sigmoid notch (olecranon + coronoid)	Trochlea
Radiohumeral (radiocapitellar)	Radial head	Capitellum
Proximal radioulnar	Lesser sigmoid notch (distal on proximal ulna)	Radial head

- Ulnohumeral joint: sigmoid notch of proximal ulna ↔ trochlea of distal humerus
  - Notch formed by olecranon + coronoid → **greater sigmoid notch**
- Radiocapitellar joint: radial head ↔ capitellum
- Proximal radioulnar joint: **lesser sigmoid notch** ↔ radial head

Interdigitation of Articular Surfaces Ask AI

- Unique articular property → one of the most stable joints
- Elbow = **highly constrained** joint
- Property = **interdigitation** of articular surfaces
  - Coronoid / olecranon against trochlea
  - Radial head against capitellum
  - Alternating concave → convex → concave segments
  - Each convexity faces a concavity, and vice versa
- Result → articular surfaces interlocked
  - Joint more stable, highly constrained

Distal Humeral Fossae and the Fat Pad Sign Ask AI

Fossae Ask AI

Humerus

- **Coronoid fossa** → proximal to trochlea
  - Accommodates coronoid tip in flexion
- **Radial fossa** → proximal to capitellum
  - Accommodates radial head in flexion
- **Olecranon fossa** → posterior surface of distal humerus
- Accommodation → increased stability in flexion
  - Adds to anteroposterior constraint
- Fossa pathology → reduced flexion and reduced stability

Fat Pads Ask AI

- Each anterior fossa → own fat pad
  - Coronoid + radial fat pads = **anterior fat pad**
- **Posterior fat pad** → in olecranon fossa
- Normal: fat pad in place, capsule lax
  - X-ray: dark line close to humeral shaft

Fat Pad Sign Ask AI

- Joint **effusion**, any type
  - Post-traumatic → blood
  - Infection → pus
  - Chronic inflammation → synovial fluid
  - Other causes
- Effusion → capsule distended anteriorly and posteriorly
  - Fluid displaces fat pad → dark line moves away from bone
- Radiographic result = **fat pad sign**
- Fat pad sign = effusion in joint
  - Cause determined from **history**
  - Examination
  - **Aspiration**, depending on situation
- Fat pad sign ≠ fracture
- Present with fractures
  - Paediatric **supracondylar fracture** of humerus → anterior and posterior sign
  - **Radial head fracture**
- Fat pad sign, no fracture visible → search for cause of effusion
  - Traumatic → additional views
    - **Oblique views**
    - Radial head views
    - Olecranon views
  - No **history of trauma** → define nature of effusion
    - **Aspiration** → synovial fluid analysis → diagnostic work-up

> **Pitfall**
> - Fat pad sign = effusion, not fracture
> - Traumatic sign with normal X-ray → **oblique, radial head, olecranon views**
> - Atraumatic sign → aspiration and synovial fluid analysis
> - Never overlook the sign

Summary of First Anatomy Image Ask AI

- **Interdigitation** of ulnohumeral and radiohumeral convexities and concavities
  - → increased stability
- Radial, coronoid fossae anteriorly; olecranon fossa posteriorly
  - Problems → limited flexion, extension and stability
  - Fractures may involve them
- Fat pads inside fossae → sign of distended joint with **effusion**
  - Effusion type worked up afterwards

Radiocapitellar Arthritis Ask AI

Congruity and Arthritis Site Ask AI

RC joint

- Trochlear **sulcus** (groove) → concave
- Adjacent segments → convex
- Olecranon, coronoid surfaces → same pattern
- Interdigitation → **joint congruity**
- Convex rim of radial head ↔ groove between trochlea and capitellum
- **Arthritis** possible at this location
  - Result → **radial-side elbow pain**

Differential Diagnosis of Lateral Elbow Pain Ask AI

- Other causes of radial-side elbow pain
  - **Tennis elbow**
  - Instability
  - Nerve entrapment
- Radiocapitellar arthritis → include in **differential diagnosis**
  - Tools: history, examination, radiographs

Imaging and Arthroscopy Ask AI

- Radial head **erosion** → lower image
- Arthritis / erosion on X-ray → may or may not be visible
  - **Isolated radiocapitellar arthritis** → may not appear on X-ray
  - Mistreated as tennis elbow, entrapment → pain persists
- **Chronic lateral elbow pain**, undetermined cause or resistant to treatment
  - Solution → **elbow arthroscopy**
  - Goal → detect intra-articular pathology
- Arthroscopic findings
  - Capitellum → erosion, articular surface not intact, chondral defect
  - Radial head → articular surface affected at contact zone
  - White articular cartilage absent
  - Surprise → **ulnohumeral joint intact**
- Arthritis between radial head and capitellum at a specific location
  - Cause → interdigitation, marked articular congruity

Literature: Isolated Radiocapitellar Arthritis Ask AI

- Radiocapitellar joint arthrofibrosis (arthritis) → links anatomy to clinical cases
- Slide purpose: where, why, best diagnosis, literature
- **2021 paper** → radiocapitellar arthroplasty and radiocapitellar arthritis
- Isolated pathology or part of generalised arthritis
- Most isolated cases → **ulnohumeral joint intact**
- Symptomatic cases → X-ray may not show arthritis
  - Most cases → arthritis not seen
- Chronic lateral elbow pain + normal X-ray → higher modality
  - **CT**
  - MRI
  - Arthroscopy → if symptomatic despite treatment

> **Clinical Pearl**
> • Chronic lateral elbow pain + normal X-ray → think **radiocapitellar arthritis**
> • Isolated cases → ulnohumeral joint usually intact
> • Work-up → CT, MRI or arthroscopy if resistant

Summary of First Two Anatomy Slides Ask AI

- **Trochoginglymoid joint** → three separate joints
- **Interdigitation** and high congruity → increased stability
- Clinical importance of fat pads
- Clinical importance of coronoid, radial and olecranon fossae

Ulnohumeral Joint and Elbow Stabilizers Ask AI

- Ulnohumeral joint: greater sigmoid notch ↔ trochlea
  - Notch formed by olecranon + coronoid
- Ulnohumeral joint = **primary bony stabilizer** of elbow

General Stability Rule Ask AI

- Every joint → primary and secondary stabilizers

Stabilizer	Primary	Secondary
Bony	Ulnohumeral joint	Radiocapitellar joint
Soft tissue	Ligaments: medial and lateral groups	Ligaments + muscles (dynamic stabilizers)

- Bony stabilizer
  - Primary → **ulnohumeral joint**
  - Secondary → **radiocapitellar joint**
- Soft-tissue stabilizer
  - Primary → **ligaments**, medial and lateral groups
  - Secondary → ligaments + muscles
    - Muscles = **dynamic stabilizers**
    - Muscles crossing in front, at sides and behind joint
  - All contribute to joint stability

Articular Contact Ask AI

- **180 degrees** of articular contact throughout arc of motion
  - Full flexion, full extension, 90° flexion → same 180°
  - 180° = half a circle → stability

Articular Geometry of the Distal Humerus Ask AI

Inclination of Articular Surfaces Ask AI

- Built-in inclination on humeral and ulnar sides
- Distal humeral articular surface → 30° to long axis
  - **Volar angulation** of 30°
- Proximal ulnar articular surface (greater sigmoid notch) → 30° dorsal
- Humeral 30° volar + ulnar 30° dorsal → reciprocate
  - Result → free motion to **full flexion** and full extension

Centre of Rotation Ask AI

- Flexion–extension **centre of rotation** → centre of articular surface
  - Located in volar-inclined anterior part
- Not on long axis of humerus
  - Shifted **anterior** to humeral long axis
  - Centre of capitellum and centre of trochlea
- Slide summary
  - Distal humerus → volar inclination
  - Proximal ulna → dorsal inclination
  - Centre of rotation → anterior

Concentric Circles on Lateral X-ray Ask AI

- Lateral X-ray → overlap of three circles

Circle	Structure
Smallest	Trochlear sulcus (narrowest part)
Intermediate	Capitellum
Largest	Medial ridge of trochlea (widest)
Additional	Greater sigmoid notch

- Smallest → **trochlear sulcus**, thinnest part of distal humeral articular surface
- Slightly larger → **capitellum**
- Largest → **medial ridge** of trochlea
- All circles → one centre = centre of rotation
- Additional circle → **greater sigmoid notch**
  - Same centre of rotation
- Centre of rotation = centre of articular arcs on X-ray

Centre of Rotation in Three Planes Ask AI

Plane	Relation	Angle
Lateral	Offset: anterior (and inferior) to long axis	4–8° (about 8°)
Axial	Internal rotation vs intercondylar axis	3–5°
Coronal	Valgus	4–8°

- Lateral plane → centre anterior and inferior to long axis = **offset**
  - About 8°
  - Range 4–8°, depending on average
  - **Arthroplasty**: correct offset → preserved range of motion and stability
- Axial plane (viewed from distal) → axis of rotation ≠ **intercondylar axis**
  - 3–5° of **internal rotation**
  - Relevant to arthroplasty
- Coronal plane
  - 4–8° of **valgus**
- Offset → anteroinferior

Carrying Angle Ask AI

- 4–8° articular valgus → contributes to **carrying angle**
  - Clinical angle between arm and forearm
- Females → larger, up to 16–18°
- Males → smaller *[C1 · transcript: "carrying angle smaller in females"]*
- 16–18° > 4–8° → additional valgus in greater sigmoid notch
- Distal humeral valgus + greater sigmoid notch valgus
  - → carrying angle = **humeroulnar angle**
  - Measured on X-ray

180 Degrees of Articular Contact Ask AI

- **180°** contact: greater sigmoid notch ↔ trochlea
- → stability throughout range of motion
  - Especially flexion → coronoid in **coronoid fossa**
- Flexion stability requires coronoid accommodation in fossa

High Yield

- Distal humerus → 30° volar; greater sigmoid notch → 30° dorsal
- Centre of rotation → **anterior** to humeral long axis
- Offset 4–8°; internal rotation 3–5°; valgus 4–8°
- Concentric circles: sulcus < capitellum < medial trochlear ridge
- 180° articular contact throughout arc of motion

Supracondylar Malunion and Distal Humeral Inclination Ask AI

Loss of Inclination: Extension Malunion Ask AI

- Normal distal humerus → **volar inclination** of 30°
- **Supracondylar fracture** fixed without restoring angle
  - Extension or excess flexion → **malposition** / **malunion**
- Distal fragment in extension → clinical case
  - Flexion → not beyond 90°
  - Extension → good, normal
  - Cause → **impingement** of coronoid in coronoid fossa
  - Bony deformity → flexion limited
- Loss of 10–20° of angulation → large loss of range of motion
- Treatment → correct bony deformity
- Wrong approach (contralateral side case)
  - Plate, **tension band** and screws removed
  - Coronoid and / or radial fossa **deepening**
  - Claimed improvement → hand reaches head
  - Deformity unchanged
  - Hand-to-head → **shoulder** flexion, not elbow
  - Elbow → still not beyond 90°
- Correct approach → **osteotomy**
  - Restore distal humeral angle to 30°
- Anterior angulation lost (extension) → **elbow flexion** problem

Increased Inclination: Flexion Malunion Ask AI

- Angle increased → 35, 40 or 45°
- Flexion range → good
- **Posterior impingement** → limited extension
- Wrong approach
  - **Triceps release** without bony correction
  - Elbow release for "stiffness"
- Correct approach → **osteotomy**, restore 30°
- Adjuncts when needed intraoperatively
  - **Deepening** of radial, coronoid or olecranon fossa
  - **Fat interposition** in fossae
- Basis → restore normal angle for flexion–extension

Distal humeral angle	Flexion	Extension	Mechanism	Correction
Normal 30°	Free	Free	—	—
Lost (extension malunion)	Limited (≤90°)	Good	Anterior coronoid impingement	Osteotomy to 30°
Increased (35–45°)	Good	Limited	Posterior impingement	Osteotomy to 30°

Pitfall

- Malunion → correct **bony angle**, not implant or fibrosis
- Implant removal / fossa deepening alone → deformity persists
- Triceps or elbow release alone → extension not restored
- Hand-to-head may come from shoulder flexion

Complex Deformities and Pseudo-extension Ask AI

- Children and others → additional **varus** or valgus
- **Complex deformity** components
  - **Sagittal plane** → flexion or extension of distal fragment
  - **Coronal plane** → varus or valgus
  - **Rotational** deformity
    - e.g. internal rotation
- **Measure** angles before correction
  - X-ray may falsely suggest extension
  - Example → good 30° articular inclination
  - Bone looks bent, as if extended → **pseudo-extension**
  - Unmeasured → unnecessary flexion correction → **limitation of extension** mid-range
- Pseudo-extension → no limitation of flexion or extension
  - Free range of motion; angle correct
- True extension → **limitation of flexion**
- Flexion deformity → limitation of extension

Finding	Pseudo-extension	True extension
Articular inclination	Correct (30°)	Lost
Range of motion	Free	Flexion limited
Action	No sagittal correction	Measure and correct

- Clinical examination
  - Free range of motion → **pseudo-extension**
  - Limited flexion → **true extension**
    - Measure angle
- Complex deformity → correct only what is indicated
- Unindicated correction → effects appear immediately **postoperatively**

Carrying Angle Measurement Ask AI

Carrying

- Clinical: long axis of arm vs long axis of forearm
  - Average **15°**
  - Smaller in males; larger in females
- Radiographic equivalent → **humeroulnar angle**
- Measurement → elbow **fully extended**
- Abnormal carrying angle → varus or valgus
  - Varus and valgus = **3D deformities**
- Case: distal humeral axis down, articular inclination lost
  - Angle → 0
  - Clinical → limitation of flexion
- Pseudo-extension → no limitation; bone only appears extended
- **True extension** → always affects flexion
- Correction components
  - Flexion (sagittal) deformity
  - **Rotation**
  - Varus / valgus
- Case: varus + extension + limitation of flexion
- Learned: distal humeral inclination → importance, measurement, consequences, decisions
- Method → **surgical anatomy**

Distal Humerus Morphology: Two Columns Ask AI

- Humerus → **cylindrical** proximally → flattens distally
- Distal humerus → **two columns**
  - Lateral column
  - Medial column
- Between columns → articular surface: capitellum + trochlea
- Fossae → olecranon fossa posteriorly, two anteriorly
  - → thinning of distal bone
- **Lateral column** wider than medial column

Column	Width	Plate option
Lateral	Wider	Dorsal plate, no articular or fossa impingement
Medial	Narrower	Contoured plate on medial border; dorsal plate → fossa impingement

- Lateral column → **dorsal plate** possible
  - No impingement on articular surface or olecranon fossa
  - Indication → distal humerus fracture
- Medial column → no dorsal plate
  - Plate → impingement in fossa

Trochlear Sulcus and Screw Placement Ask AI

Trochlea

- Trochlear groove → narrowest part of articular surface
- Medial / lateral plate screws or **interfragmentary screws** → centre of trochlea
  - Too proximal → **impingement** in coronoid or olecranon fossa
  - Too distal → damage to trochlear groove and articular surface
- Two columns: medial narrower than lateral
  - Lateral → **dorsal plate**, avoid impingement
  - Medial → **contoured plate** on medial border
- Modern **anatomical plates**
  - Incorporate all these features

Exam Pearl

- Distal humerus → two columns; **lateral wider** than medial
- Lateral column → dorsal plate
- Medial column → contoured plate on medial border
- Screws → centre of trochlea, avoid fossae and groove

Proximal Ulna Anatomy Ask AI

- Proximal ulna → important to study
  - Basis of **primary stabilizer**
- Olecranon + coronoid → **greater sigmoid notch**
- Greater sigmoid notch arc → **190°**
- **Lesser sigmoid notch** ↔ radial head
  - → proximal radioulnar joint
- Coronoid → **anteromedial facet**
  - Tubercle on facet → **sublime tubercle**

Sublime Tubercle and Anterior Band of UCL Ask AI

- Name → **flexor digitorum sublimis** (superficialis)
  - Ulnar head originates from sublime tubercle
- Sublime tubercle + ulnar ridge
  - → attachment of **anterior band** of ulnar collateral ligament
  - Most important ligament in ulnar collateral complex

Supinator Crest and LUCL Ask AI

- Lesser sigmoid notch
  - Articulates with radial head
  - Ridge = **supinator crest** → origin of supinator
- Proximal part of crest → radial-side ligament attachment
  - **Lateral ulnar collateral ligament (LUCL)**
  - Most important lateral ligament

Landmark	Attachment	Significance
Sublime tubercle (anteromedial facet)	FDS ulnar head; anterior band of UCL	Most important medial ligament
Supinator crest (proximal part)	Supinator; LUCL	Most important lateral ligament

Fractures at Ligament Attachments Ask AI

- Broken anteromedial facet or sublime tubercle on X-ray
  - = attachment of **primary ligament** of medial stability
  - Not simple → major component of stability
  - Anteromedial facet large → part of ulnohumeral joint
  - Ulnohumeral joint = primary bony stabilizer
  - → **reconstruct** or repair for stability and function
- Lateral side → same when fracture reaches **bony attachment** of ligament
  - Same principle in humerus at ligament attachments

Must-Know

- Sublime tubercle → anterior band of **UCL**
- Supinator crest → LUCL
- Anteromedial facet fracture → repair or reconstruct for stability

Non-articular Area and Notch Arcs Ask AI

- Greater sigmoid notch → **non-articular** line between coronoid and olecranon
  - Clinical use → discussed later
- Greater sigmoid notch → about 190°
- Lesser sigmoid notch → **70°**
  - Measured from centre of radial head circle
- 190° → more stability than 70°

Greater Sigmoid Notch: Facets and Load Sharing Ask AI

Gliding Ridge Ask AI

Ulna

- Recall: **interdigitation**, congruity, highly constrained
- Parts of greater sigmoid notch
  - Posterior **capsule**
  - Olecranon side; coronoid side
  - Central ridge → **gliding ridge**
- Gliding ridge → **convex**
- Ridge divides articular surface → medial + lateral parts
- **Medial part larger** than lateral part
- Medial part attachments
  - Anteromedial facet
  - Capsule
  - **Medial collateral ligament**
- Capsule surrounds whole joint

Medial Facet: Resisting Varus Ask AI

- Ulnohumeral joint = primary bony stabilizer
- Medial part larger
  - → greater stress and strain than lateral part
- Medial facet → resists **varus**
  - In extension and early flexion
- Medial facet → mainly responsible for varus stability
- Helpers
  - **Lateral collateral ligament**, opposite side
  - Joint capsule
- Varus stress → resisted by medial part of **ulnohumeral joint**
  - Large part → carries larger load
- Varus → lateral side stressed → **lateral ligament** resists
- Extension → capsule shares → **three structures**
- Flexion beyond 30° → capsule drops out
  - No varus / valgus stability from capsule

Lateral Facet: Resisting Valgus Ask AI

- Lateral part → not truly small
- Cooperates with **radial head** → resists valgus
- Radial head → increases its area
- Valgus load distributed
  - Lateral part
  - Radial head
  - **Capsule**
  - Ligament
- Lateral part role → proportional to size

Load-sharing Percentages Ask AI

Stress	Position	Ulnohumeral facet share	Other contributors
Varus	Extension	Medial facet 55%	LCL + capsule 45%
Varus	Flexion	Medial facet 75%	Capsule drops out
Valgus	Extension	Lateral part 30%	Radial head, capsule, MCL
Valgus	Flexion	Lateral part 30%	Capsule load → radial head + MCL

- **Varus in extension** → medial facet bears most
  - Medial facet → **55%**
  - Remaining 45% → lateral collateral ligament + joint capsule
  - Largest share → medial part of ulnohumeral joint
- **Varus in flexion** → capsule eliminated
  - Medial share → **75%**
- Explains larger area of medial part of greater sigmoid notch
- Lateral part → resists **valgus**
  - Valgus in extension → **30%**
  - Valgus in flexion → **30%**
  - Flexion: capsule load → **radial head** + medial ligament
  - **Medial collateral ligament** deficient → load mostly to radial head
- MCL problem → preserve **radial head**
  - Radial head carries stress → no longer secondary stabilizer
- Slide lesson
  - Interdigitation, perfect congruity
  - Articular area reflects load carried
  - Each surface shape → specific helping structures
- Further detail later in course

High Yield
- Varus in extension → medial facet **55%**; LCL + capsule 45%
- Varus in flexion → medial facet 75%
- Valgus → lateral part 30% in extension and flexion
- Capsule → no varus / valgus role beyond 30° flexion
- MCL deficiency → radial head primary → preserve it

Olecranon Osteotomy Ask AI

- Basis → **non-articular** part between coronoid and olecranon
- Indication → **dorsal (posterior) approach** for distal humeral articular exposure
  - **Olecranon osteotomy**
- Articular surface spared

  - → through non-articular part

Technique Ask AI

Osteotomy

- Dorsal approach
- Isolate **ulnar nerve**
- Open joint → pass **dissector**
- Dissector passes opposite **non-articular surface**
  - → level of **osteotomy**
  - Apex of chevron rests on it
  - Protects olecranon and coronoid articular surfaces
- Start with **saw** → complete with osteotome
- Pre-drill screw hole before osteotomy
  - Fixation ready at end
- Design from anatomy → non-articular part located by dissector
- Details → separate session
- Olecranon reflected
  - Whole distal humerus exposed
  - **Posterior fat pad** and olecranon fossa visible

Fixation Case: Supracondylar Fracture Ask AI

- **Supracondylar fracture** → olecranon osteotomy + fixation
- Older plating; now **anatomical plates**
  - Cut plate → **posterolateral**
  - **Medial plate** on medial border after contouring
- **Transverse screws** → through narrowest point
  - Point not clearly visible → easily passed blindly
  - Common trouble point during reduction → fractures session
- Anatomy study → creative surgical ideas

> **Clinical Pearl**
> - Dissector through opened joint → marks **non-articular** level
> - Chevron apex at this level → articular surfaces protected
> - Saw first, osteotome to finish
> - Pre-drill fixation screw hole

Proximal Ulna Angles Ask AI

Angle	Normal	Reported range	Reference
Proximal ulna dorsal angulation (PUDA)	5–7°	0–14°	Contralateral side
Proximal ulna varus angulation	Slight varus	—	Contralateral side

Proximal Ulna Dorsal Angulation (PUDA) Ask AI

PUDA

- **PUDA** → about 5–7°
  - Proximal ulna not straight → volar inclination
  - Dorsal border not parallel to proximal shaft
- Measurement: line on dorsal border vs line on proximal shaft
  - Range 5–7°
- Reference → **contralateral side**
  - Reported range 0–14°
  - Risk: set to 0 or 14° when patient's original was 5–6°
  - Patient compares result with healthy arm
- Any ulnar fracture → measure PUDA on both sides

Proximal Ulna Varus Angulation Ask AI

- Distal ulnar axis ≠ proximal axis
  - Angle → ulna in slight **varus**
- Limits → compare with **contralateral side**

Distal Radius Analogy Ask AI

- Unrestored angle → consequences
- Distal radius → **volar inclination** about 12°
- Dorsally displaced fracture → **dorsal inclination** of 10°
  - Volar 10° → dorsal 10°, or restored only to zero or less
- Displacement → at expense of **distal radioulnar alignment**
- Dorsal angulation with full pronation and supination → needs study
  - Alignment depends on two bones: radius displaced, ulna in place
  - → rotation limited
- Same principle proximally
  - PUDA (volar inclination) and varus
  - Unrestored → affects **proximal radioulnar joint**
  - Key to **Monteggia** lesions

Monteggia Lesions Ask AI

Plastic Deformation Ask AI

Monteggia

- Child with **plastic deformation** → no obvious fracture
  - Ulnar dorsal angulation changed
  - At expense of **proximal radioulnar joint** → dislocated
  - Line drawn → dislocation shown
- Any fracture → check **proximal radioulnar joint** dislocation
  - Easily missed
  - Error: "simple plastic deformation" → **splint** only
  - Reduction needed to restore PRUJ
  - Missed → **neglected** case → late dislocation

Pitfall

- Plastic deformation of ulna → check **PRUJ**
- Splint alone without reduction → neglected Monteggia

Neglected Monteggia: Kocher Approach Ask AI

- Lateral **Kocher approach**
  - Interval: **anconeus** ↔ extensor carpi ulnaris
  - Other lateral approaches → indications discussed later
- Findings
  - **Lesser sigmoid notch** empty
  - Radial head dislocated
- Reduction without angle correction → redislocation
  - Unstable
  - Open reduction + (annular) ligament **repair** alone → redislocates
  - **K-wire** fixation → redislocates after removal
- Solution → correct proximal ulnar angle
  - Angle visible on previous X-ray
  - **Osteotomy** → then reduction
  - Then address ligaments
- **Annular ligament** often not torn → pulled / displaced
  - Sometimes torn
  - Both → repair / reconstruct
  - Does not return exactly → avoid over-tightening

Monteggia in Adults Ask AI

- Proximal ulnar anatomy → can cause **Monteggia** lesion
- Monteggia → occurs in adults
- Plate well placed, small angle unnoticed
  - Detected later by function
  - Limited motion
  - **Instability** of proximal radioulnar joint
  - Arthritis, pain

Must-Know

- Neglected Monteggia → **ulnar osteotomy** before radial head reduction
- Ligament repair or K-wire alone → redislocation
- Annular ligament often displaced, not torn
- Monteggia occurs in adults too

Closing Remarks Ask AI

- Next sessions → expanded **Monteggia** discussion
  - Focus → **paediatrics**
  - Management, reasoning, avoiding missed diagnosis
  - Builds on ulnar angulation and varus
- Anatomy → basis of every clinical situation
- Comments and corrections welcome

Elbow Course Part 2: Surgical Anatomy of the Proximal Radius

(Radial Head, Neck and Bicipital Tuberosity)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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C2 (p. 38) “flexion weakness 50% (also said 30%)” → “flexion strength decreases ~30%”

Queried - kept as spoken, but looks wrong

Q1 (p. 43) As spoken; 360° minus 250° articular is 110°, so 120° does not add up.

Proximal Radius: Overview Ask AI

- Previous lecture → bony surgical anatomy of **distal humerus** and proximal ulna
- This lecture → **proximal radius**
- Three parts
  - **Radial head**
  - **Radial neck**
  - **Radial tuberosity**
- Radial head articulations
  - **Capitellum**
  - **Lesser sigmoid notch** of proximal ulna
- Role → elbow and forearm motion
  - Flexion, extension, pronation, supination
- Important component of the **elbow joint**
- Key features → tuberosity, neck, head

Bicipital (Radial) Tuberosity Ask AI

Anatomy Ask AI

Tuberosity

- Site → **posteromedial aspect** of proximal radius
- Two parts
  - Anterior → smooth
  - Posterior → rough → **posterior ridge**

Biceps Attachment Ask AI

Tuberosity

- Biceps tendon attachment → closely related to tuberosity
- **Anatomical studies** → no direct attachment on tuberosity or ridge
- Attachment area → **posterior (ulnar) to the ridge**

Function of the Ridge Ask AI

Tuberosity

- Smooth anterior part → **prominence**
  - Tendon glides over it; may protect tendon from attrition
- Prominence → **mechanical pulley**, not merely a facing surface
  - ↑ **mechanical advantage** (tendon power) of biceps
  - Mainly in **supination**; also in flexion

Must-Know

- Ridge → **not the direct site** of attachment
- Ridge → **mechanical pulley** → biceps advantage in forearm supination
- Relevant → **distal biceps tendon tear** repair or reconstruction

Tuberosity Morphology: 2018 Cadaveric Study Ask AI

- 2018 → study of bicipital tuberosity and biceps tendon
- **Cadaveric anatomical study** → 178 radii
- Three tuberosity forms
  - **Smooth** → no ridge
  - **Single ridge**
  - **Bifid** → two ridges
- Single ridge → **posterior**
- Bifid → effectively two ridges
- Single ridge size → **small / medium / large**, measured in mm

Types and Frequencies Ask AI

Tuberosity

Type	Features	Frequency
Smooth	No ridge at all	6% — uncommon
Bifid	Anterior part + posterior ridge part	6% — uncommon
Single ridge	Smooth anterior part; posterior ridge — small, medium, large	88% — largest proportion

- Images → from **original paper**

Smooth Type Ask AI

Tuberosity

- Smooth → predisposes to **tear or avulsion**? → no association found
- Idea → no ridge → altered **supination** and forearm rotation
  - Clinical problem → **needs further study**
- Bifid → anterior part + posterior ridge part → 6%
- Purpose of such studies → new findings that aid **diagnosis**
- Predominant type → **single ridge**

Single Ridge Type Ask AI

Tuberosity

- Anterior part smooth → posterior part carries **ridge**
- Ridge height
  - **Small** → low
  - **Medium** → moderate
  - **Large** → great

Large Ridge — Clinical Relevance Ask AI

Tuberosity

- **Large ridge** → 12% of the 88% single-ridge tuberosities
- Pronation → large ridge **narrows radioulnar space**
  - → **attrition** or attenuation of biceps tendon
  - → tear or degeneration
  - → possible **predisposing factor** for distal biceps tear (author's hypothesis)
- Author's key finding → biceps tendon **not attached** onto ridge
- Tendon passes over ridge → inserts on **ulnar portion**, dorsal/posterior to it
- Ridge → **bony pulley** → ↑ tendon mechanical advantage

Surgical Implications for Re-attachment Ask AI

Tuberosity

- Distal biceps re-insertion
  - Do not **remove the ridge**
  - Do not **shorten the tendon**
  - Do not attach onto the ridge
  - Fix tendon **dorsal to the ridge** → mechanical advantage preserved
- Author → points must be respected to benefit from **re-attachment**

High Yield

- Do not **remove the ridge**
- Do not attach tendon **onto the ridge**
- Attach **dorsal to the ridge** → mechanical advantage preserved

Distal Biceps Tendon Tear: Clinical Features Ask AI

- Clinical relevance of tuberosity anatomy → **distal biceps tendon tear**
- Not common

Mechanism of Injury Ask AI

DBT tear

- Usual → **forced extension** with elbow flexed
  - Elbow flexed → forced extension → tendon **tears or avulses**
- Opposite → lifting heavy object from extension into flexion
  - Possible, but not common
- Common → **sudden extension** against forced flexion
- Uncommon → tear while **lifting a heavy object**
  - → pre-existing tendon problem
    - **Degeneration**
    - **Attenuation**
    - **Corticosteroid** use

Clinical Presentation Ask AI

DBT tear

- History → **mechanism of injury**
- **Pain** and swelling
- **Ecchymosis** around elbow
- Muscle belly **retracted proximally** → gap between belly and elbow
- **Elbow crease** → absent
- → **reverse Popeye sign**

Popeye vs Reverse Popeye Ask AI

DBT tear

Sign	Biceps belly	Lesion
Popeye (original)	Distal migration	Proximal biceps tear
Reverse Popeye	Proximal retraction	Distal tendon tear

Chronicity Ask AI

DBT tear

- Assess → distance between **muscle belly** and elbow crease
  - Large distance → probably **chronic** (case shown)
  - Small distance → **subacute or acute**, especially with ecchymosis

Clinical Tests for Distal Biceps Tear Ask AI

Hook Test Ask AI

DBT tear

- Very important
- Elbow flexed → 90°
- **Index finger** → hooks biceps tendon
- **Sensitivity** → ~100%
- Self-test → flex elbow, hook tendon → tendon felt
- Torn tendon → hook **fails to catch** tendon
- **False-negative** → torn, yet tendon seems felt *[C1 · transcript: "hook test gives false positive"]*
  - **Partial tear**
  - Tight **lacertus fibrosus** (bicipital aponeurosis, arising from biceps tendon)
- Otherwise → **reliable**

Squeeze Test Ask AI

DBT tear

- Elbow flexed → 60–70°
- **Squeeze biceps** muscle → forearm supination
- Intact tendon → **supination** produced
- Analogous → **Thompson test** of Achilles tendon
  - Squeeze **gastrocnemius** → plantar flexion if tendon intact
  - Same principle → distal biceps tendon

Hook vs Squeeze Ask AI

DBT tear

	Hook test	Squeeze test
Elbow flexion	90°	60–70°
Manoeuvre	Index finger hooks tendon	Squeeze biceps muscle
Intact tendon	Tendon caught	Forearm supination

- **Hook test** → sensitivity and specificity ~100%

Pitfall
- **Partial tear** → both tests misleading
- Strong **lacertus fibrosus** → tendon presence unclear

Imaging in Distal Biceps Tear Ask AI

- Equivocal tests → another **diagnostic modality**

Modality	Findings
Ultrasound	Tear present or not; dynamic assessment in flexion, extension, supination; gap size; lacertus vs tendon itself
MRI	Further clarification; gap; chronic or not

- **X-ray** → like salt in food → every case
  - Usually **normal**
  - Some reported cases → **avulsion** (bony fragment) from radial tuberosity

Surgical Approaches to the Radial Tuberosity Ask AI

- Surgical anatomy → reaching tuberosity for **re-attachment**
- Three approaches
  - **Anterior** → elbow opened anteriorly
  - **Posterior** → between anconeus and ulna
  - **Combination** → double approach
- Each → own advantages and disadvantages

Anterior Approach Ask AI

Tuberosity

- Long approach → expose **biceps tendon** and lacertus fibrosus
- Structures at risk
  - **Brachial artery**
  - **Median nerve**
  - **Posterior interosseous nerve (PIN)**
  - **Radial nerve**
  - **Lateral antebrachial cutaneous nerve** → termination of musculocutaneous nerve; very important
- Careful dissection before tuberosity reached → **time-consuming**
- Access to dorsal ridge for **tunnel** → cramped
- Disadvantages
  - **Time-consuming**
  - Injury risk → **PIN** and lateral antebrachial cutaneous nerve

Posterior (Boyd) Approach Ask AI

Tuberosity

- Two-site entry considered
- Posterior entry between anconeus and ulna → **Boyd approach** → easier access to radius

Treatment of Distal Biceps Tendon Tear Ask AI

Conservative Ask AI

DBT tear

- Options → conservative or operative
- Indications
  - **Unfit for surgery**
  - **Old age**, low demand, satisfied with function
- Measures → **supportive measures** and physiotherapy
- Inform patient
  - **Flexion strength** ↓ ~30% *[C2 · transcript: "flexion weakness 50% (also said 30%)"]*
  - **Supination strength** ↓ ~50% → more important than flexion
- Copes well → no problem

Operative Ask AI

DBT tear

- Not coping; needs supination and flexion power; active arm → **re-attachment**, repair or reconstruction
- Reaching tuberosity for re-insertion → two techniques
  - **Single (anterior) approach**
  - **Double approach** → two incisions

Single Anterior Approach Ask AI

DBT tear

- **Longitudinal** or transverse incision
- Longitudinal → explore tendon and **lateral antebrachial cutaneous branch**
  - Of musculocutaneous nerve → avoid cutting it
- Ligate vessel **leash**
- Explore **PIN**
- Reach **radial tuberosity** → tunnel → all work anteriorly
- Risks → lateral antebrachial cutaneous nerve or **PIN** injury
- Dissection → **time-consuming**

Double Approach Ask AI

DBT tear

- **Anterior incision** → transverse (case shown)
- **Posterior incision** → marked over radial head and lateral condyle
- Steps
  - Transverse incision → identify, protect **lateral antebrachial cutaneous branch** (musculocutaneous termination)
  - Retrieve **biceps tendon** → prepare tendon and sutures
  - **No deep anterior** dissection to tuberosity
  - **Mosquito** forceps, blunt dissection → tip felt posteriorly over planned incision
  - → **blunt tunnel** → tendon passed to posterior incision
  - **No entry** between ulna and anconeus
  - Forceps emerges posteriorly through **extensor digitorum** / extensor compartment → cut down onto mosquito

Why Not Between Anconeus and Ulna Ask AI

DBT tear

- Problem → **anconeus elevated**
  - Entry between anconeus and ulna → **periosteal stripping** of ulna during tuberosity work
- Both together → heterotopic ossification or radioulnar **synostosis**
- → avoid anconeus–ulna interval; **preserve ulna**
- Stay away from ulna → **complication avoided**

Summary of Approaches Ask AI

DBT tear

Approach	Route	Drawback / advantage
Anterior	All work anterior	Long duration; cutaneous nerve and PIN injury risk
Double via Boyd	Anterior + posterior, anconeus elevated off ulna	High risk → ulnar periosteal stripping → synostosis
Double via mosquito	Anterior + dorsal cut-down onto mosquito	No extensive anterior dissection; no anconeus elevation

- Mosquito emerges **dorsally** → away from ulna
- Cut down onto mosquito → dorsal side → **tunnel** → anchors, sutures → repair
- Avoids **extensive anterior** dissection and its time
- Avoids anconeus elevation and ulnar **periosteum** stripping → no synostosis risk
- Preferred by most surgeons → **distal biceps tendon repair** or reconstruction

Clinical Pearl

- Never enter between **anconeus and ulna** → synostosis risk
- **Mosquito tunnel** → dorsal cut-down → ulna preserved
- Most surgeons' preferred technique for **distal biceps** repair

Radial Neck–Shaft Angle Ask AI

Radial Neck Ask AI

Neck

- **Radial neck** → between radial tuberosity and radial head
- Key feature → **inclination** → radial neck–shaft angle

Values Ask AI

Neck

Source	Neck–shaft angle
Commonly quoted	~15°
Paper discussed	Mean 7° ± 3° (SD); range 0–14°

- Measure **bilaterally**, as with other angles
  - Patient's angle → **contralateral side**; population ranges less relevant

Why It Matters Ask AI

Neck

- **Fracture** needing fixation → accurate reduction → angle (15°) preserved
  - → **pronation and supination**
  - → prevents later **arthritis**
- Correct angle → restores forearm **alignment**, stability, rotation (pronation, supination)
- Confirm → compare **other side**
  - 15° and ranges → **cadaveric studies** and population data
  - Individual patient → own **contralateral side**

2025 Study: Lateral Elbow Pathology Ask AI

Neck

- ~2025 study → angle vs **lateral epicondylitis** and LCL degeneration; relevance to treatment
- Question → **chronic lateral epicondylitis** + LCL injury/degeneration correlated with neck–shaft angle?
- Hypothesis → **larger** proximal radial neck–shaft angle associated?
- Design
  - **Retrospective study** → 57 cases
  - **Radiologists** → reviewed MRI
  - **Shoulder and elbow surgeons** → examined patients
  - **Resistant lateral epicondylitis** + LCL degeneration → angle measured
  - **Corticosteroid** recipients excluded → avoid bias
    - Steroid injection for lateral epicondylitis → itself causes **LCL degeneration**
  - Angles measured → **X-ray** and MRI
- Result → **increased angle** correlated with lateral epicondylitis and LCL degeneration
- Author's hypothesis → needs **further research**
- Main message → **anatomical reduction** → rotation, stability, no arthritis
- Inadequate reduction, ↑ angle → **LCL degeneration** and instability (author) → interlinked

Prosthesis Design Ask AI

Neck

- Second application → **prostheses** incorporate neck angle → proximal radius anatomy preserved

Radial Head: Shape and Stability Ask AI

Shape Ask AI

Radial head

- **Articular**, disc-shaped, not round
- **Ellipsoid** → AP diameter ≠ transverse diameter
- Non-circular → advantage in **rotation** (pronation, supination)
  - Circular head → would behave differently

Valgus Stability Ask AI

Radial head

MCL status	Radial head share of valgus stress	Role
Intact	30%	Secondary stabilizer
Deficient	75%	Primary stabilizer

- Applies to valgus in **extension or flexion**
- MCL problem → radial head → **much greater importance**
- Instability, elbow trauma, **radial head fractures** → excision no longer accepted
  - **Reconstruct/repair** bone → restore it
  - **Arthroplasty** (prosthesis) → if irreparable

High Yield
- Intact MCL → radial head bears 30% valgus → **secondary stabilizer**
- Deficient MCL → 75% → **primary stabilizer**
- **Radial head excision** → no longer accepted

Load Transmission Ask AI

Radial head

- 60% of axial force → **radiocapitellar joint**
- Carrying or leaning on hand → force through **ulna and radius**
  - **Radius** → carries 60% → elbow via radial head

Longitudinal Forearm Stability Ask AI

Radial head

- Radial head → **longitudinal stability** of forearm
- Radial head fracture → **DRUJ** instability
- **Essex-Lopresti** injury
  - Radial head fracture
  - DRUJ disruption
  - **Interosseous membrane** disruption
- Longitudinal forearm stability → radial head as **secondary stabilizer**
- Radial head → secondary stabilizer of **DRUJ** itself
- DRUJ stabilizers
  - **Anteroposterior**
  - **Transverse**
  - **Longitudinal** → radial head, proximally

Radial Head: Articular Arcs and Safe Zone Ask AI

Articulations Ask AI

Radial head

- Ellipsoid → **AP and transverse** diameters differ
- **Lesser sigmoid notch** → 70° arc of the 360° circle
- **Articular cartilage** → 250°
- 70° → lesser sigmoid notch at **any rotation** (neutral, pronation, supination)
- Remainder of 250° → **annular ligament**
- **Fovea centralis** → concave → articulates with convex capitellum
- 360° – 250° articular → **remainder**

Safe Zone Ask AI

Radial head

- Remainder → 110–120° **non-articular** → safe zone *[Q1 · As spoken; 360° minus 250° articular is 110°, so 120° does not add up.]*
- Use → **fracture fixation**, e.g. plate
- Image → **fovea centralis** → depression articulating with capitellum
- Image → **lesser sigmoid notch** articulation → 70° arc

Arc of 360°	Articulation
70°	Lesser sigmoid notch
Rest of 250°	Annular ligament
110–120°	None → safe zone

- Plate on radial head or neck → **safe zone**
  - → full rotation, pronation, supination postoperatively
  - No plate **blocking** pronation

Intraoperative Identification Ask AI

Safe zone

- **Articular cartilage width** → clearly greater in one region than anterolateral part
- → **intraoperative sign** of safe zone
  - Safe zone cartilage → never reaches **lesser sigmoid notch** in pronation or supination
  - **Wide cartilage** → articulates with lesser sigmoid notch
  - Look for sign while operating
- After exposure → **rotate forearm**
  - Wide, large vs thin, small articular surface
  - Thin part → **plate site**
- Anatomical line → neutral rotation → safe zone **lateral**
- Line aligned with safe zone and **lateral condyle** (capitellum)
  - Full supination and pronation → yellow line moves forward and backward
  - Never enters **articular part** of lesser sigmoid notch
  - No block to pronation or supination → **plate on this line**
- Also described → between **radial styloid** and Lister's tubercle, extended proximally
- Intraoperatively → identified by **articular surface width**
  - Wide on one side; thin, small on other → thin = safe zone → **plate**

Must-Know

- Safe zone → 110–120° **non-articular** arc
- Neutral rotation → safe zone **lateral**
- Between radial styloid and **Lister's tubercle**, extended proximally
- Intraoperative sign → **thin articular cartilage**

Posterior Interosseous Nerve and the Radial Head Ask AI

- Structure of greatest concern in radial head approach → **PIN**
- **Posterior interosseous nerve (PIN)**
  - Supplies **dorsal (extensor) compartment** of forearm, with exceptions below
  - Runs **close to radial head** and neck
- Central to **exploration** and surgical anatomy of radial head and neck

Extensor Innervation Ask AI

PIN

Muscle	Nerve
Brachioradialis	Radial nerve
Extensor carpi radialis longus	Radial nerve
Remaining extensors	Posterior interosseous nerve

- **Brachioradialis** → highest origin from distal humerus

Course of the Radial Nerve Ask AI

PIN

- **Groove** → between brachioradialis and brachialis (with biceps); brachialis beneath
- **Radial nerve** → runs in this groove
- At **lateral epicondyle** → terminal branches
  - Supplies brachioradialis and **extensor carpi radialis longus**
  - Continues as **radial nerve** (likely the superficial branch)
  - Gives **PIN** → passes dorsally beneath supinator (probably at arcade of Frohse)
    - Under cover of **extensor carpi radialis brevis**
- → **close relation** of nerve to radial head
- Important and challenging → **surgical exploration** of radial head, neck, tuberosity
- Next lecture → **approaches to radial head and neck** in detail

Elbow Course Part 3: Lateral Surgical Approaches

(Approaching the Radial Head and Neck from the Lateral Side)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Queried - kept as spoken, but looks wrong

Q1 (p. 58) As spoken; lateral epicondylitis is provoked by resisted wrist extension (Cozen), not resisted wrist flexion.

Q2 (p. 58) As spoken; release attributed to 1979 here but to 1972 later in the same lecture (Roles and Maudsley 1972).

Q3 (p. 59) As spoken; radiocapitellar capsule is normally the tunnel floor and brachioradialis/ECRB the roof, so roof/floor look reversed.

Q4 (p. 60) As spoken; lateral epicondylitis test is resisted wrist extension, not resisted wrist flexion.

Lateral Approaches to the Elbow: Overview Ask AI

- Elbow course part 3 → **surgical anatomy**: surgical approaches
  - Previous two lectures → bony anatomy: distal humerus, proximal ulna, proximal radius
- Focus → **lateral approaches**: access to radial head and radial neck
  - Clinical topics alongside anatomy → better retention
  - More than one lateral route into the elbow joint
- Best-known lateral approaches
  - **Kaplan approach** → ECRB–EDC interval
    - Lecture covers: identifying muscles, finding interval, entry, key structures
    - When to choose Kaplan over Kocher and other approaches
  - **Kocher approach** → ECU–anconeus interval
  - **Subperiosteal elevation of anconeus** → access from posterior
  - **Extensile approach** → lateral humeral epicondyle osteotomy
    - Epicondyle elevated with all attached structures → lateral access to joint

Surface Anatomy and Lateral Muscle Order Ask AI

Surface Landmarks Ask AI

- Upper figure → muscle anatomy and surface landmarks of lateral elbow
  - **Lateral humeral epicondyle**
  - **Radial head**
  - Triangle (**lateral soft spot**) → joint injection / aspiration
- **Lateral intermuscular septum** + **supracondylar ridge** of distal humerus
  - Lateral humeral epicondyle
  - Septum → origin of **forearm extensor muscles**

Brachioradialis Ask AI BR

- First, most superior origin on supracondylar ridge → **brachioradialis**
  - Insertion → radial styloid
  - Nerve → **radial nerve**
- Radial nerve → under cover of brachioradialis
  - Groove between brachioradialis and **brachialis**
- This muscle group → **mobile wad**
- Interval opened like a book → radial nerve between brachioradialis and brachialis
- Brachioradialis → **first muscle to receive radial nerve branch**
  - After nerve passes lateral intermuscular septum / spiral groove

Radial Nerve Recovery Test Ask AI

BR

- Palsy at spiral groove or more proximal → test **brachioradialis** first
  - First supplied after septum / spiral groove → first to recover
- Elbow 90 degrees, mid-pronation → **resisted elbow flexion**
  - Muscle felt contracting → recovery occurred / muscle working

Clinical Pearl

- Radial nerve palsy at or above spiral groove → **brachioradialis** recovers first
- Test → resisted elbow flexion, elbow 90 degrees, mid-pronation

ECRL and Anconeus Ask AI

- **Extensor carpi radialis longus (ECRL)** → second after brachioradialis
  - Origin → lateral supracondylar ridge
  - Insertion → base of second metacarpal
  - Nerve → main radial nerve, before division into superficial radial nerve and **PIN**
- **Anconeus** → last muscle here; radial nerve
- Radial nerve → **three muscles**
  - Brachioradialis → most superior origin
  - ECRL
  - Anconeus
- Remaining forearm and finger extensors → **posterior interosseous nerve**

High Yield

- Radial nerve → brachioradialis, ECRL, anconeus
- All other forearm and finger extensors → **PIN**

EDC and ECRB Ask AI

Kaplan

- Third in order → **extensor digitorum**
- ECRB → not between lateral origins; comes from distal
  - Lies **deep to EDC**
  - Passes deep to ECRL and EDC → reaches its origin
- **Kaplan approach** → open between ECRB and EDC
  - Finding interval → go distal, identify, follow proximally
  - First trick → find interval **from distal**

ECU and Kocher Interval Ask AI

Kocher

- Posterior to EDC → **extensor carpi ulnaris (ECU)**
- ECU–anconeus interval → **Kocher approach**
  - Clearly visible from lateral → found precisely
  - **Easiest** interval
- Subperiosteal elevation of anconeus → **Boyd approach** (anconeus approach) *[C1 · transcript: "anconeus subperiosteal elevation = Kaplan approach"]*
- Whole lateral side with muscles → osteotomy, elevated → **extensile approach**
  - Indication → more comprehensive view of elbow joint

Approach	Interval / technique	Note
Kaplan	ECRB – EDC	Find interval from distal
Kocher	ECU – anconeus	Easiest interval
Boyd	Subperiosteal elevation of anconeus	Posterior access
Extensile	Lateral epicondyle osteotomy, elevated with attached muscles	Most comprehensive view

Lateral Condyle Origins and Lateral Epicondylitis Ask AI

Figure: Origins from the Lateral Humeral Condyle Ask AI

- Three figures; left figure → muscle attachments to **lateral humeral condyle**
- Supracondylar ridge (red) → two muscles
  - Brachioradialis → superior part
  - ECRL → part below
- ECRB → deep to EDC origin, covered by it
  - EDC–ECRB interval → identified from distal, beneath
  - Figure → ECRB origin, other muscle origins, deep to anconeus

Capsular Attachment Hypothesis Ask AI

LE

- Source → **paper** on muscle origins vs **capsular attachment**
- Closer origin to capsular attachment → more **degeneration** at that origin
- **Lateral epicondylitis** → degeneration in ECRB more than EDC
  - ECRB → **closest to joint capsule**
  - **W1** → ECRB–capsule distance, small
  - **W2** → EDC–capsule distance, large
- Author's hypothesis → one cause of degeneration in lateral epicondylitis

Figure: Separated Lateral Forearm Muscles Ask AI

- Lower right figure → lateral forearm muscles, separated
  - Brachioradialis
  - ECRL
  - EDC
    - ECRB → passes **deep to EDC**
    - Interval → go distal, identify, follow proximally
  - ECU
  - **Anconeus** → last; Kocher interval between anconeus and ECU
- Order from above → BR → ECRL → ECRB → EDC → ECU

Order	Muscle	Origin / position	Nerve
1	Brachioradialis	Highest on supracondylar ridge → radial styloid	Radial nerve
2	ECRL	Supracondylar ridge, below BR → base of 2nd metacarpal	Main radial nerve
3	ECRB	Deep to ECRL and EDC; closest to capsule	PIN
4	EDC	Superficial to ECRB	PIN
5	ECU	Posterior to EDC	PIN
6	Anconeus	Last; Kocher interval with ECU	Radial nerve

Supinator and PIN Deep to the Lateral Intervals Ask AI

- Muscles elevated → **supinator** beneath; **PIN** runs within
- PIN and radial neck → reachable through any interval
- Kocher interval → **pronation** protects nerve, moves it ~30 mm

Interval	Exposure
Brachioradialis – ECRL	Radial neck and PIN exploration
Brachioradialis – biceps (more anterior)	PIN and supinator over radial head
ECRB – EDC (Kaplan)	Supinator and PIN; exploration
ECU – anconeus (Kocher)	Supinator; exploration; pronation moves nerve ~30 mm
Extensile approach	Nerve explored normally

PIN Entrapment: Brachioradialis–ECRL Approach Ask AI

PIN

Case and Skin Incision Ask AI

- Clinical case → interval between brachioradialis and ECRL
  - Alternative → Kaplan approach
- Diagnosis → **PIN entrapment / radial tunnel syndrome**
  - Diagnosis discussed later → approach opened
- Approach → between brachioradialis and ECRL
- Locating brachioradialis → elbow flexed 90 degrees against resistance
- Groove just posterior to brachioradialis → between it and ECRL
  - Groove marked on skin → **skin incision**

Lateral Antebrachial Cutaneous Nerve Ask AI

- First structure sought → **lateral antebrachial cutaneous nerve**; protect
  - Termination of musculocutaneous nerve
  - Marks brachioradialis–ECRL interval
- Nerve retracted → interval between brachioradialis (red) and ECRL
  - Nerve → dark structure
- Nerve protected → marks interval

Developing the Interval Ask AI

- Brachioradialis retracted; ECRL **retracted posteriorly**
- ECRB passes deep to ECRL and EDC → seen directly beneath retracted ECRL
- ECRL superficial; ECRL and ECRB retracted
- Wound floor → between brachioradialis and ECRL
  - ECRL retracted one side, brachioradialis other side
  - Anatomy clearly seen
- **Fat streak** → contains nerve and vessels
  - No rush to look for PIN yet
- Steps so far
  - Skin incision → brachioradialis–ECRL interval identified
  - **Lateral antebrachial cutaneous nerve** protected → marks interval
  - Interval opened superficially → **ECRB in wound floor**
  - ECRB origin → deep to ECRL and EDC

Locating the PIN via the Superficial Radial Nerve Ask AI

- Fat streak → **superficial radial nerve**, covered by brachioradialis
  - Follow **proximally** → leads to **PIN**
- Search method for PIN
  - Find nerve distally → follow proximally → PIN

PIN versus Nerve to ECRB Ask AI

PIN

- Second nerve, fairly large → **nerve to ECRB**; not PIN
- Field: ECRL (white) one side; brachioradialis retracted other side
  - Floor → **ECRB** beneath ECRL
- Fat streak beneath ECRL → search for nerve
  - Nerve beneath brachioradialis → superficial radial nerve
- Proximally: nerve ≈ superficial radial nerve size or thinner → **not PIN**
- **PIN → larger than superficial radial nerve**
  - Never equal, never thinner
- Nerve to ECRB → branch near PIN

> **Pitfall**
> • Nerve equal to or thinner than superficial radial nerve → **nerve to ECRB**
> • PIN → always larger than superficial radial nerve

Vascular Leash Ask AI

- **Vascular leash** crossing → ligate and cut
  - One cause of PIN compression
  - Ligated, excised → view beneath
- Second figure: ECRL → ECRB beneath; brachioradialis → superficial radial nerve beneath
  - Superficial radial nerve followed proximally → adjacent nerve
  - Equal or thinner → **nerve to ECRB**
- Before reaching PIN → vascular leash in front → cut, separate

ECRB Origin Release Ask AI

- ECRB origin → separated from nerve, taken transversely, **excised**
  - Reason `1` → PIN **compression**
  - Reason `2` → origin **degeneration** → lateral epicondylitis; pain source with PIN
- Cutting origin → deals with both causes of pain
- Right figure: ECRL; brevis beneath; brachioradialis
  - ECRB origin opened → **fibrous band** covering whole origin taken
  - Muscle fibers left
  - **Fibrous band over supinator** → divided
- PIN appears beneath → thick, passing **beneath supinator**

Compressing Structures of the PIN Ask AI

Arcade of Frohse Ask AI

PIN

- Supinator → **arcade of Frohse** opened
  - Before opening → nerve passing beneath, compressed
  - Nerve segment under arcade → appears thinned
- Release sequence
  - Vascular leash cut
  - ECRB **fibrous origin** cut
  - Arcade of Frohse released
  - **Entire supinator opened** → distal part may compress nerve

Four Compressing Structures Ask AI

- **Four structures** compress PIN
  - **Vascular leash**
  - **Origin of ECRB**
  - **Supinator arch** (arcade of Frohse)
  - **Supinator itself** + distal fibrous band
- Supinator opened → nerve released
  - Final figure → supinator opened until full release

Must-Know

- PIN compression → vascular leash
- PIN compression → ECRB origin
- PIN compression → **arcade of Frohse**
- PIN compression → supinator and its distal fibrous band

Summary of the Approach Ask AI

- Incision → between brachioradialis and ECRL
- Beneath ECRL → nerves; beneath brachioradialis → **superficial radial nerve**
- Superficial radial nerve → adjacent nerve, equal or thinner → PIN, thicker than both
- Compressing forces on PIN
  - **Fibrous band** over radial head and capitellum → removed
  - Structures cut
  - **Supinator** may compress → must be opened
- Brachioradialis identified by origin above → BR–ECRL interval opened
- Lateral antebrachial cutaneous nerve → identifies interval, protected
  - Interval opened → final view reached
- Other procedures possible through this exposure

Prevalence of Upper-Limb Nerve Entrapment Ask AI

- Radial tunnel → described in papers as **rare**
- Paper on **prevalence of upper-extremity nerve compression** → surprising result
  - Lecturer → statements backed by papers

  - Paper presented next lecture
- Most common → **lacertus fibrosus syndrome**, not carpal tunnel

Lacertus Fibrosus Syndrome Ask AI

Lacertus

- **Lacertus fibrosus** → fibrous band at biceps; **median nerve** passes beneath
  - Compression under it → **lacertus syndrome**, most common entrapment
- Carpal tunnel presentation → exclude **proximal entrapment** at lacertus
  - Distal entrapment combined with proximal? Or lacertus syndrome alone?

Examination for Proximal Median Compression Ask AI

Lacertus

- Motor exam → **FCR** power vs other side
- **FDP index and middle** power vs other side → median nerve
- Weakness of FCR + FDP index / middle → **proximal median compression**, usually lacertus
- Percussion / tapping over lacertus at elbow → positive **Tinel sign** ± local tenderness

Rank	Entrapment
1	Lacertus syndrome
2	Carpal tunnel
Next	Radial tunnel syndrome (PIN entrapment)

- Considered rare → prevalence data say otherwise → **radial tunnel important**

Exam Pearl

- Commonest upper-limb entrapment → **lacertus syndrome**, not carpal tunnel
- Carpal tunnel case → check FCR and FDP index / middle for proximal compression

Radial Tunnel Syndrome: Definition and History Ask AI

Definition Ask AI

RTS

- Source → **2023**, American Academy of Orthopaedic Surgeons, hand subspecialty issue
  - Year corrected by lecturer from 2025
- **Entrapment neuropathy** → **chronic pain**, usually lateral forearm
- Initially → considered non-resolving lateral epicondylitis
  - Similar symptoms; diagnosis unclear
- **1956** → author: separate entity from lateral epicondylitis
  - Lateral epicondylitis → pain over lateral epicondyle, ECRB origin
  - Nerve entrapment → pain distal, about **3 cm** below
  - Site → over radial head and radial neck

Provocative Tests Ask AI

RTS

- Resisted forearm supination **with elbow extension** → pain increases
  - Supinator contracts → closes on PIN passing through it
  - Supinator → supination with elbow in extension
  - Elbow in flexion → main supinator = **biceps**
- Test 1 → **resisted supination with elbow extension**
  - Elbow extended → supinator works, closes on nerve
- Test 2 → **resisted middle-finger extension with elbow extension** (ECRB, extensor digitorum)
- Lateral epicondylitis → **resisted wrist flexion** → pain over lateral epicondyle *[Q1 · As spoken; lateral epicondylitis is provoked by resisted wrist extension (Cozen), not resisted wrist flexion.]*
- Separate entity → PIN entrapment, described 1956

Night Pain Ask AI

RTS

- Radial tunnel entrapment → **wakes patient from sleep**; lateral epicondylitis does not
  - As in carpal tunnel → major symptom, diagnostic feature
  - Ask: pain and numbness waking from sleep?

Historical Milestones Ask AI

- Second author, **1979** → nerves releasable; examination tests established *[Q2 · As spoken; release attributed to 1979 here but to 1972 later in the same lecture (Roles and Maudsley 1972).]*
- Diagnostic criteria set → **1979**
- Release for cases failing medical treatment → **1972**
- Three stages

Year	Milestone
1956	Lateral epicondylitis distinguished from radial tunnel (PIN) entrapment
1972	**Release** proposed
1979	Diagnostic tests distinguishing it from lateral epicondylitis

- 1972 release → cases not improving with **medical treatment**

Radial Tunnel Anatomy and Surgical Access Ask AI

Boundaries Ask AI

RTS

- Tunnel → boundaries, roof, floor
- Radial nerve descends under **mobile wad**: brachioradialis, ECRL, ECRB
  - Opening between them and brachialis → radial nerve
- Radial nerve **divides ~3 cm** distal to lateral humeral epicondyle
- Radial nerve beneath brachioradialis
  - Posterior (interosseous) branch → through **supinator** → posterior compartment
- Figure → lateral and medial boundaries
- Dark structure → biceps and brachialis (second boundary)

Boundary	Structure
Lateral	Brachioradialis, ECRL
Medial	Biceps tendon, brachialis
Floor	ECRB lying on nerve; supinator
Roof	Radial head, radial head capsule *[Q3 · As spoken; radiocapitellar capsule is normally the tunnel floor and brachioradialis/ECRB the roof, so roof/floor look reversed.]*

- Space so defined → **radial tunnel**

Surgical Access Ask AI

RTS

Route	Note
Anterior → **Henry approach**	Familiar approach to proximal forearm
Brachioradialis – ECRL	As in case shown
ECRB – EDC (Kaplan)	Yes
ECU – anconeus	Yes; the turn not far

- Tunnel reachable from any of these intervals

Radial Tunnel Syndrome: Diagnosis Ask AI

Clinical Features Ask AI

RTS

- Pain increasing; may wake from sleep
- **No motor weakness**
  - EMG and nerve conduction → no motor weakness
  - Confusing → **high level of suspicion** needed
- Clinical tests
  - Resisted supination with elbow extension
  - Resisted middle-finger extension with elbow extension

Differentiation from Lateral Epicondylitis Ask AI

- Lateral epicondylitis → pain over lateral epicondyle
- Radial tunnel → pain over radial head / radial neck, distal
- **2 fingerbreadths** distal to lateral humeral epicondyle → pain site
  - **Safe zone** → lateral epicondyle to PIN crossing = two fingerbreadths
  - Site of PIN pain → differs from lateral epicondylitis
- Lateral epicondylitis test → **resisted wrist flexion** elicits pain

Feature	Radial tunnel syndrome	Lateral epicondylitis
Pain site	Radial head / neck; ~3 cm, 2 fingerbreadths distal	Lateral epicondyle, ECRB origin
Provocative test	Resisted supination / middle-finger extension, elbow extended	Resisted wrist flexion *[Q4 · As spoken; lateral epicondylitis test is resisted wrist extension, not resisted wrist flexion.]*
Night pain	Wakes from sleep	Does not
Lesion	PIN entrapment	ECRB origin degeneration

Investigations Ask AI

RTS

- **Ultrasound** → useful with obvious cause
  - Lipoma, ganglion, nerve schwannoma, any large mass
  - Otherwise → nerve diameter and similar measurements
- **MRI** → nerve diameter, thickness
- Neither conclusive; EMG / nerve conduction not conclusive
- Some EMG studies → may be detectable
- Despite **neuropathic** nature → EMG and NCS **negative** in suspected cases
- Diagnosis → **clinical sense** and **clinical examination**

Pitfall

- Radial tunnel syndrome → no motor weakness
- EMG and nerve conduction → negative in suspected cases
- Diagnosis rests on **clinical examination**; high suspicion needed

Radial Tunnel Syndrome: Treatment Ask AI

Conservative Treatment Ask AI

RTS

- Start → **conservative treatment**
- **Analgesics** or **steroid injections** → follow up result
- **>50%** resolve with conservative treatment
  - Diagnosis alone → not indication for surgery
  - Operate only if no resolution

Injection Ask AI

RTS

- **Corticosteroid + local anesthetic** → possible **temporary PIN palsy**
  - Injected around nerve, not into it
  - **Ultrasound-guided** → mandatory; otherwise refer
  - Palsy from local anesthetic → recovers
- Another author → steroids give no results
  - Yet many cases resolve
- **Resistant cases → surgical release**

Clinical Pearl

- >50% of radial tunnel cases → resolve conservatively
- Steroid + local anesthetic → temporary PIN palsy, recovers
- Injection → around nerve, **ultrasound-guided** only

Radial Tunnel Syndrome: Diagnostic Criteria Ask AI

- **Thorough history**
  - Key question → **does it wake you from sleep?**
- **Examination** → site of pain; diagnostic tests
- **Mechanical allodynia** → after brief work, hand tires, gives way; pain increases

  - **Weakness with work**, not pain alone
  - Felt especially in motor nerve entrapment, e.g. PIN
- PIN → motor to entire posterior forearm compartment except **3 muscles**

Approaches for PIN Decompression Ask AI

- Final figure → **ECRB origin**
- **Supinator** beneath
- Anterior opening or any intermuscular interval → reaches **supinator**
- Author's approaches → **PIN decompression**
- Entry options
  - Anterior approach
  - Brachioradialis – ECRL
  - **Kaplan approach**

Next Lecture Ask AI

- Continued approaches → **Kaplan**, **Kocher**, **Boyd**
- Kaplan → advantages, exposure, disadvantages; differences between approaches

Elbow Course Part 4: Lateral Approaches 1 - Kaplan Approach

(Indications, Interval and Risk to the Posterior Interosseous Nerve)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Lateral Surgical Approaches: Indications Ask AI

- Part 4 of surgical anatomy → **surgical approaches** → lateral approaches first
- Indications for a lateral approach
  - Proximal radius fractures
    - **Radial neck** fracture
    - **Radial head** fracture
  - Distal humerus fractures
    - **Capitellar fracture**
    - **Lateral condyle fracture**
    - **Column fracture**
    - **Trochlear fracture** → also accessible laterally
  - Capitellar **osteochondral defect** requiring treatment
  - **LCL insufficiency** → repair or reconstruction
  - **Lateral epicondylitis** → when open procedure planned
  - **Capsular contracture** (stiff elbow) → open release

Patient Positioning Ask AI

- **Supine** → standard position, used by most surgeons
- Arm options
  - **Hand (arm) table** → elbow extended, shoulder abducted
  - **Elbow flexed 90°** → forearm across chest
    - Surgeon standing; no side table needed
- Key points
  - Intraoperative imaging → shift patient laterally
    - Toward table edge → easy **image intensifier** movement
    - AP and lateral views obtained without difficulty
  - Risk → **traction injury** to nerves around shoulder
    - **Suprascapular nerve**
    - **Axillary nerve**
    - Prevention → **bolster under the shoulder** → stops traction on arm
      - No traction injury to shoulder nerves

Pitfall

- Supine, arm positioning → **traction injury** to suprascapular or axillary nerve
- Bolster under shoulder → removes traction on arm

Overview of the Lateral Approaches Ask AI

- Several lateral approaches, anterior → posterior

Approach	Interval / technique
Kaplan	ECRB ↔ extensor digitorum
Extensor digitorum splitting	Extensor digitorum itself split
Kocher (more posterior)	ECU ↔ anconeus
Boyd (posterior)	Anconeus elevated off ulna → posterior joint entry

- Two additional approaches
  - Exposure of **lateral column** of distal humerus
  - **Lateral epicondyle osteotomy** → whole joint, including medial access, via lateral approach

Kaplan Approach: Indications, Advantages, Disadvantages Ask AI

- Kaplan approach → first lateral approach covered

Indications Ask AI

Kaplan

- **Radial head** or radial neck fractures
- **Anterolateral distal humerus** fractures or pathology

Interval Ask AI

Kaplan

- **ECRL and ECRB** ↔ extensor digitorum
- Interval identification → covered in earlier lecture

Advantage Ask AI

Kaplan

- Lies anterior to and away from **LUCL** → low ligament risk
  - Cutting or injuring **LUCL** → practically absent, as long as you stay anterior
  - Anterior to LUCL **humeral attachment**
  - Anterior to LUCL **ulnar attachment**
  - → low risk of ligament tear

Disadvantage Ask AI

Kaplan

- **Close proximity to PIN**
  - PIN protection → **pronate the forearm** well
  - Large proximal radius exposure → **PIN exploration** preferred

Kaplan	Detail
Advantage	Anterior to LUCL → ligament protected
Disadvantage	Close to PIN → nerve at risk
Protection	Forearm pronation; PIN exploration for wide exposure

Skin Incision and Superficial Dissection Ask AI

Skin Incision Ask AI

Kaplan

- Landmarks → **Lister tubercle** + lateral epicondyle
- Line: lateral epicondyle → Lister tubercle
  - Incision along line → **6–8 cm**
  - Proximal point → **lateral humeral epicondyle**
  - Distal point → **Lister tubercle**
  - Start at lateral humeral epicondyle → 6–8 cm distally
- Incise skin → **subcutaneous tissue**
  - Kaplan slightly anterior → **lateral antebrachial cutaneous nerve** encountered
  - Path not empty → protect nerve branch in elevated skin flap

Exam Pearl

- Kaplan incision → lateral epicondyle to **Lister tubercle**, 6–8 cm
- Nerve at risk in skin flap → **lateral antebrachial cutaneous nerve**

Superficial Dissection Ask AI

Kaplan

- Skin flaps raised → anterior and posterior
- Lateral muscles, proximal → distal
  - **Brachioradialis** → most proximal, from supracondylar ridge of distal humerus
  - **ECRL** → below it
  - **ECRB** → common extensor origin, lateral humeral epicondyle
- ECRB **undercover** → beneath ECRL and extensor digitorum → not visible
  - On opening → ECRL visible, extensor digitorum next to it
- Entry → **extensor digitorum ↔ ECRB**
  - Find interval **distally** → two muscles clearly distinct there
  - Then follow interval **proximally**
  - ECRB ↔ extensor digitorum = **Kaplan interval**
- Interval → anterior to **LUCL attachment**
  - LUCL and ligament complex → attach to **lateral epicondyle**
  - Interval anterior to ligament → **ligament injury** usually avoided

High Yield

- Kaplan interval → **ECRB** (with ECRL) ↔ extensor digitorum
- Find interval distally → follow proximally
- Anterior to LUCL → ligament spared

Deep Dissection: Supinator and PIN Ask AI

- Interval EDC ↔ ECRB (with longus) opened
- Floor → **supinator muscle**
- Supinator contains **PIN** → volar, within its fat pad, in muscle substance
- PIN protection → **forearm pronation**

  - Pronation → PIN shifts
    - **Volarly**, away from field → ~4 cm gained
  - Without pronation → ~2.5 cm
  - With pronation → ~4 cm proximally via this approach

Forearm	Safe distance
Not pronated	~2.5 cm
Pronated	~4 cm

- Disadvantage → **PIN at risk**
- Many surgeons avoid this approach → regarded as largely **historical**
  - Possibly reserved for interval already opened by injury; closed interval → avoid Kaplan
  - Alternative → shift posteriorly to another approach

Must-Know
- Supinator in floor → contains **PIN**
- Pronation → PIN moves volarly → ~4 cm vs ~2.5 cm

Technique Summary and Illustrations Ask AI

Steps Ask AI

Kaplan

- If Kaplan unavoidable
  - Open **EDC** ↔ **ECRB** → find interval distally
  - **Supinator** → pronate → elevate **subperiosteally** anteriorly
    - PIN protected by pronation
  - Radial exposure to **6 cm** → **explore the nerve**
    - → protect PIN, avoid injury

Cadaveric and Clinical Illustrations Ask AI

Kaplan

- Cadaveric elbow picture + patient picture
- Incision → **lateral epicondyle** to distal Lister tubercle
  - ~6–8 cm, or as much as needed
  - Procedure not needing 6–8 cm → shorter acceptable
- Interval between
  - **Extensor digitorum** → posterior
  - ECRL and ECRB → anterior
- Floor → **supinator**; PIN volar
- Pronation → more space
- Interval → extensor digitorum ↔ **extensor carpi radialis**
  - Not brevis alone → ECRB together with longus

Fascia Technique Ask AI

Kaplan

- Distal end of approach → open **fascia** with scalpel → small opening
- Extend with **scissors** → proximal and distal
- Not all with scalpel → **snip** with scalpel → scissors up and down
- **Slip muscle off fascia** → open fascia
- Enter → extensor digitorum ↔ ECRB and ECRL
- Retract extensor digitorum and ECRL/ECRB → **supinator** in floor

Supinator and Retractors Ask AI

Kaplan

- Supinator → contains **PIN**; figure = Kaplan approach
- Pronate → elevate supinator **subperiosteally**
- Retractors on medial (volar) side → **forbidden**
  - **Hohmann retractor** → posterior only
  - Any retractor → never volar

Hohmann placement	Verdict
Posterior	Allowed
Medial (volar)	Forbidden

Capsulotomy and the Lateral Ligament Complex Ask AI

- Skin incision → extensor digitorum ↔ ECRB and ECRL
- Open → **supinator** → **joint capsule** ahead
- Capsule opening options → volar, middle, dorsal
  - Correct → **middle, at the equator**
    - Pronation or supination → open capsule at midpoint
- **Capsule** + annular ligament exposed
- Kaplan → anterior to **LCL attachment**
  - LCL attaches → **lateral humeral epicondyle**
  - LCL components
    - **Lateral radial collateral ligament**
    - **LUCL**
    - **Annular ligament**
- Capsule opened together with **annular ligament**
  - Rest of complex → posterior → untouched, unreachable
- **LUCL injury** needing repair or reconstruction → Kaplan not suitable
  - Ligament dorsal to you → cannot be reached
  - Ligament work → **different approach** required
  - Kaplan → passes anterior to ligament, no cut → advantage

Clinical Pearl

- LUCL repair or reconstruction → **Kaplan not suitable**
- Capsulotomy → **equator**, midpoint of capsule

Kaplan

Summary Ask AI

- Exposes **radius** + distal humerus, slightly anterior
- Capsule opened at **equator**, middle
- Pronate → **supinator** elevated subperiosteally
- More exposure → **explore PIN** → extend proximally

Structures at Risk Anteriorly Ask AI

- Very important point → structure running with
  - Brachioradialis (probable) and **biceps** → runs down along them
- **Hohmann retractor** placed medially →
  - Possible injury to radial nerve (PIN per take-home points)
  - Injury to **radial recurrent artery**

Take-Home Points: Kaplan Approach Ask AI

- Only advantage → **protects lateral ligament complex**
  - LCL repair planned → do not use Kaplan
- Exposes **anterior joint** without problem
- Problem → **PIN at risk**
  - Prevention → pronate forearm + **subperiosteal elevation** of supinator
- No **Hohmann retractor** medially
  - Risk → traction injury to **PIN**
  - Risk → injury or tear of **radial recurrent artery**

High Yield

- Kaplan → only advantage: **lateral ligament complex** protected
- Kaplan → main risk: **PIN**
- Medial Hohmann → PIN traction, radial recurrent artery injury

Elbow Course Part 5: Modified Kocher Approach

(The Anconeus - Extensor Carpi Ulnaris Interval)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 80) As spoken; lateral distal humeral surface (capitellum) is convex; the radial head surface is the concave one.

Q2 (p. 81) As spoken; earlier the lecture gives pronation PIN shift as ~1.5–2 cm, not 2–2.5 cm.

Modified Kocher Approach: Overview Ask AI

- Context: elbow course → surgical anatomy → **lateral approaches**
  - Previous lecture → Kaplan approach
  - This lecture → **modified Kocher approach**
- **Most familiar** elbow approach; most surgeons work through it
- Scope: advantages + associated risks
- Interval: **anconeus** ↔ extensor carpi ulnaris (ECU)
- Structure at risk: **posterior interosseous nerve (PIN)**
  - Protect during both distraction and retraction

High Yield

- Modified Kocher interval → **anconeus / ECU**
- Structure at risk → **PIN**

Nerve Safe Zones Around the Lateral Epicondyle Ask AI

- **Pronation** → larger PIN safety margin during distraction
- Landmark: **lateral humeral epicondyle** → mark, then measure
  - 4 fingerbreadths proximal → **radial nerve** risk point
    - Nerve crosses **lateral intermuscular septum** here
  - 2 fingerbreadths distal → **PIN** risk point

Distance from lateral epicondyle	Nerve at risk	Note
4 fingerbreadths proximal	Radial nerve	Passes posterior → lateral in arm through lateral intermuscular septum
2 fingerbreadths distal	Posterior interosseous nerve	Shifts further distal with pronation

- **Pronation** → PIN risk point shifts distally ~1.5–2 cm
- Kaplan, modified Kocher, all lateral approaches → **pronate the forearm**
  - Moves PIN away from distraction field

Lateral Collateral Ligament Access and Indications Ask AI

Why Kocher Reaches the LCL Ask AI

Kocher

- Kaplan: dissection more **anterior**
  - Extensor-to-extensor → **LCL** posterior to field → unreachable
- **Fracture-dislocation** → expected LCL tear → Kaplan cannot reach → Kocher
- Planned **LCL repair or reconstruction** → Kocher
  - Better access
- Flip side: no LCL injury → risk of cutting **ligament** or its attachment
  - Working very close to it

Capsulotomy Level Ask AI

Kocher

- Capsulotomy: midway, on **equator of radial head**
  - Volar (anterior) → **PIN**
  - Dorsal (posterior) → **lateral ulnar collateral ligament (LUCL)**

Indications Ask AI

Kocher

- **Radial head fractures**
- **Capitellum fractures**; trochlea also reachable
- **Lateral epicondyle** pathology
- **Ligament repair** or reconstruction

Skin Incision Ask AI

- Mark **lateral humeral epicondyle**
- Mark ulnar point 6 cm distal to **olecranon**
- Connect the two points
- Modified Kocher incision: lateral epicondyle → **ulna** 6–8 cm distal to olecranon
- Differs from **Kaplan incision**
  - Kaplan: also starts at lateral humeral epicondyle
  - Line between landmarks; length possibly ~6–8 cm (unclear in recording)
- Kaplan interval anterior to **LCL** → no access → repair/reconstruction needs modified Kocher

Feature	Kaplan	Modified Kocher
Interval	Extensor-to-extensor, more anterior	Anconeus / ECU
Relation to LCL	Anterior to LCL → cannot reach it	Better access to LCL
Fracture-dislocation with LCL tear	Not suitable	Approach of choice
LCL repair / reconstruction	No	Yes

Superficial Dissection Ask AI

- Anconeus/ECU interval → usually **clearly visible**
- Technique:
  - **Snip** in the fascia
  - Open fascia with **scissors**
  - Separate the two muscles with scissors
  - Avoid straight scalpel cut top → bottom
- Snip fascia → separate fascia from muscle → open → **dissecting scissors** between ECU and anconeus
- Figure landmarks:
  - **Lateral humeral epicondyle**
  - Ulnar point ~6–8 cm from **tip of olecranon**
  - **Anconeus**
  - **ECU** → fiber direction slightly different from anconeus
- Scalpel snip → fascia off muscle → **dissection**

Deep Dissection and Capsulotomy Ask AI

- Anconeus and ECU separated from each other
- Floor, proximal part of incision: **supinator** + elbow capsule
- Open capsule at **equator** (midline)
  - Avoid anterior and posterior
  - Posterior → **LCL** + LUCL
  - Anterior → **PIN** risk
- Enter center of **radial head**
  - Head round in pronation or supination → aim for middle of circle

Capsulotomy direction	Structure at risk
Anterior (volar)	Posterior interosseous nerve
Posterior (dorsal)	LCL, LUCL
Equator of radial head	Safe entry

- After capsulotomy → **radial head**, capitellum, radiocapitellar joint clearly visible

Retractors and Distal Extension Ask AI

Hohmann Retractor Placement Ask AI

Kocher

- **Hohmann retractors** → posterior, never anterior
- First retractor → **ECU**
- Second retractor → **ECU + capsule**
- Anterior Hohmann → PIN or **recurrent radial artery** injury
  - Hence posterior Hohmann

Extending the Incision Distally Ask AI

Kocher

- **Supinator** lies distally
- Steps:
  - **Pronate** the forearm
  - Possible extension → ~2 cm
  - Beyond → **explore the PIN** to avoid cutting it

Clinical Case 1: Exposure and Radial Head Fracture Ask AI

Skin and Superficial Exposure Ask AI

Case 1

- Cases operated through the **Kocher approach**
- Marked landmarks: olecranon, **lateral humeral epicondyle**, ulna
  - Ulnar point from olecranon tip → joined to lateral epicondyle
- **Skin + subcutaneous tissue** → opened as one layer
- Anterior risk in subcutaneous dissection → **lateral antebrachial cutaneous nerve**
  - Slightly anterior; keep in mind
- Two muscles + **interval** clearly seen
  - **Anconeus** posterior; ECU anterior
- **ECU** with the extensor muscles
- Open interval: scalpel **snip** → fascia off both muscles → scissors
- Anconeus posterior; ECU + **extensor group** anterior → interval → capsule
- **Supinator** in this region → contains PIN

Supinator and Distal Limit Ask AI

Case 1

- Retraction: anconeus posterior; ECU + extensor group anterior; **joint capsule** in front
- **Supinator** → marked by yellow lines
  - Seen → **pronate** forearm → extra PIN protection
  - During capsulotomy → distal extension into supinator stays **safe**
- Limit: ≤2–2.5 cm (~two fingerbreadths) distal to **lateral humeral epicondyle**
  - **Pronation** → adds roughly another half
  - Further into supinator → **identify the PIN**

Capsulotomy Findings Ask AI

Case 1

- Interval: anconeus; extensor carpi radialis/ulnaris + extensor group; opened **capsule**
- Finding → **radial head fracture**
  - Fragment **impacted** and inverted (flipped); rest of radial head adjacent
  - Capsule shown in green → **edges** of capsulotomy
- Capsule opened at **equator**
  - Neither anterior nor posterior
- **Supinator** visible; no opening beyond capsule needed
- **LUCL** not opened

Lateral Collateral Ligament Repair Ask AI

Lateral Ulnar Collateral Ligament Ask AI

LUCL

- **LUCL** → main stabilizer of lateral complex → do not cut
  - Check: torn → repair; intact → no repair needed
  - Cut during approach → **repair it**
- Unnoticed cut or forgotten repair → postoperative elbow **instability**
  - Likely posterolateral rotatory type (unclear in recording)
  - Instability caused by the approach itself
- This case, after radial head work → inspect **ligament**
  - **Footprint** on lateral epicondyle empty → ligament avulsed
  - Distal attachment → **supinator crest** of ulna
  - Main lateral stabilizer → **must be repaired**

Pitfall

- LUCL cut unnoticed or left unrepaired → postoperative **elbow instability**
- Intact LUCL → no repair; cut or torn → repair

Anchor Placement Ask AI

Case 1

- Lateral epicondyle = site of avulsed ligament → reattach with **anchor**
- Anchor → **most proximal point** of lateral humeral epicondyle, not center
  - Tensioning → ligament lies in center of **footprint**
  - Anchor in middle → tensioned ligament lies **distal**, below footprint
  - Otherwise → ligament not lying correctly
- Here: avulsed ligament → anchor at most proximal point → **repair**

Suture Technique Ask AI

Case 1

- Anchor sutures → ligament + **extensor group**, anteriorly and posteriorly
  - Continue along capsule distally
- Suture handling:
  - One limb → anterior side; other limb → posterior side
  - **Tie limbs together** → ligament lies down
    - With muscle mass on lateral humeral epicondyle = **common extensor origin**
- Repair continued along **capsule** with muscle mass
- Tighten proximally → continue distally → final view
- Final view:
  - Extensor muscles + **supinator**
  - **ECU** + extensor group
  - **Anconeus**
  - Extensor mass + capsule + ligament + anchor → **lateral complex** with common extensor origin
- Wound closed

Video Demonstration Ask AI

- Procedure on **video**:
  - Mark incision → open
  - Reach interval → separate muscles
- Muscles separated → deep structure exposed
- Goal → **preserve the ligament**; inspect before closure
- Case → **radial head fracture**
- Distal extension → no scalpel; scalpel only at **ligament level**
  - **Step by step**, not all at once
  - Expose working level; all in **pronation** → away from PIN
- Then address the pathology
- Radial head fractures → later in course
  - **Classification** and management
  - Arthroscopic vs open: when, and advantages of each
  - Equivalent, or one superior
- Radial head → **preliminary fixation**
  - → **definitive fixation**
- Partial video only (full video longer); focus on surgical approach

Clinical Case 2: LCL Avulsion in a Child Ask AI

Imaging and Approach Choice Ask AI

Case 2

- Patient → **child**
- Imaging:
  - **X-ray** → radial head fracture only
  - **CT** → avulsion fracture of lateral humeral epicondyle
    - → **LCL avulsed** from its attachment
- Approach → **modified Kocher** → better lateral ligament view than Kaplan

Exposure and Fixation Ask AI

Case 2

- **Anconeus** posterior; ECU + extensor group anterior
- Capsule opened → radial head fixed with **plate**
- Lateral humeral epicondyle → ligament **avulsed**
- **K-wire** here → only marks the physis
  - Avoids mistaking physis for ligament **avulsion site** from this angle
  - True avulsion site → more **proximal**
- Ligament returned to its site
- Child → **physis**, not avulsion area; some confuse the two

Pitfall

- Child: **physis** ≠ ligament avulsion site
- K-wire marks physis; true avulsion site more proximal

Anchor Repair Under Tension Ask AI

Case 2

- **Anchor proximally**
  - From most proximal point → ligament lies on footprint
  - Anchor in middle → ligament off footprint when tensioned
- Applies to **every ligament** repair
  - E.g. metacarpophalangeal ligament, **gamekeeper's** (thumb ulnar collateral) ligament
  - Anchor in center of footprint
    - → tensioned ligament misses footprint → anchor at **proximal edge**
- Sutures tied over **anchor** → lateral ligament reattached under tension
- **Muscle mass** closed with capsule + lateral ligament, all under tension
  - Soft tissue mass relaxed in these cases → repair **tension** restores elbow stability

Clinical Pearl

- Anchor at **proximal edge** of footprint, not center
- Tensioning then seats ligament on its footprint
- Applies to every ligament repair, e.g. MCP, gamekeeper's

Intraoperative Stability Assessment Ask AI

- **Intraoperative X-ray** after plate fixation of radial head
  - **Fracture-dislocation** → joint was dislocated
- Reduction check:
  - **Articular surfaces**: concave distal humeral surface laterally + radial articular surface *[Q1 · As spoken; lateral distal humeral surface (capitellum) is convex; the radial head surface is the concave one.]*
  - **Joint space** medial vs lateral → distance equal
  - **No discrepancy**; discrepancy → residual instability
  - **Congruent** joint: no medial or lateral widening → good stability
- Plus **clinical testing** → test stability intraoperatively in instability cases
  - Fracture-dislocation, after ligament repair → pronation, supination, flexion, extension
  - Unstable in supination → retest in **pronation**

Intraoperative finding	Action
Stable	Done
Unstable in supination, stable in pronation	Immobilize in pronation
Unstable in supination and pronation	Address medial compartment + medial collateral ligament

Summary Ask AI

- Approach easy, without problems
- Technical points:
  - Ligament tear → **modified Kocher** → better ligament view
  - No ligament tear → avoid **cutting the ligament**

  - Capsulotomy at **equator of radial head** → neither anterior nor posterior
  - **Hohmann** posterior; anterior → PIN traction injury or recurrent radial artery injury
    - Anatomy → see previous lecture
  - **Pronate** forearm → more distance for distal extension onto proximal radius
    - Pronation → additional 2–2.5 cm *[Q2 · As spoken; earlier the lecture gives pronation PIN shift as ~1.5–2 cm, not 2–2.5 cm.]*
  - **LCL repair** with extensor muscle mass
    - Never ligament alone; never dissected free from extensor mass
- Later in course → **elbow instability**, fracture instability

Must-Know

- Interval → **anconeus / ECU**; risk → PIN
- Capsulotomy → **equator of radial head**
- Hohmann → **posterior** only
- LCL repaired with **extensor mass**, anchor at proximal edge
- Forearm **pronated** throughout

- Next lecture: third lateral approach → **EDC splitting approach**
  - Combines advantages of Kaplan + modified Kocher; avoids disadvantages of both in relevant cases

Extensor Digitorum Communis Splitting Approach

(Sparing the Posterior Interosseous Nerve and the Lateral Ulnar Collateral Ligament)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Overview and Rationale Ask AI

Course Context Ask AI

EDC

- Elbow course → previous lectures: **Kaplan approach** → Kocher approach
- Today: **extensor digitorum communis (EDC) splitting approach**

Rationale Ask AI

EDC

- Aim → avoid or reduce disadvantages of Kaplan and Kocher approaches
- **Kaplan approach**
  - Interval: EDC ↔ extensor carpi radialis longus (ECRL) + brevis (ECRB)
  - Somewhat anterior → **posterior interosseous nerve (PIN)** at risk
    - During traction and exploration
- Dissection moved dorsally → through **EDC muscle itself**
  - → reduced risk to PIN
  - PIN still needs care during traction and exploration
- **Kocher approach**
  - Interval: **extensor carpi ulnaris (ECU)** ↔ anconeus
  - Somewhat dorsal → **lateral collateral ligament complex** at risk
  - Proximally → dissection lies on LCL attachment
    - Especially **lateral ulnar collateral ligament (LUCL)**
  - LCL injury → a Kocher disadvantage
- Dissection moved anteriorly → safer for **lateral collateral ligament**

Advantages Retained Ask AI

EDC

- Kaplan: anterior → good view of **radial head**, coronoid, capitellum
  - Retained with EDC splitting
- **Modified Kocher approach** major advantage → LCL attachment visible
  - Tear or injury → identify → repair or reconstruct
  - Also possible through EDC splitting
- Many surgeons prefer **EDC splitting** over Kaplan and Kocher
  - Unless case needs an extension different from both

Approach	Interval	Structure at risk	Advantage
Kaplan	EDC ↔ ECRL + ECRB (anterior)	PIN	Radial head, coronoid, capitellum view
Modified Kocher	ECU ↔ anconeus (dorsal)	LCL complex, especially LUCL	LCL attachment seen → repair or reconstruct
EDC splitting	Through EDC itself	PIN risk reduced; LCL safer	Both advantages retained

High Yield

- Kaplan → **PIN** at risk
- Kocher → **LCL complex** (LUCL) at risk
- EDC splitting → less PIN risk + LCL safer
- Keeps Kaplan anterior view + Kocher **LCL access**

Skin Incision and Superficial Dissection Ask AI

Skin Incision Ask AI

EDC

- Same as **Kaplan approach** incision
- Proximal landmark: **lateral humeral epicondyle**
- Distal landmark: probably **Lister's tubercle** → line between = incision
- Length: 5 to 6–7 cm from lateral epicondyle along line
- Skin + subcutaneous tissue opened → flaps raised → muscles exposed
- Anterior flap → **lateral antebrachial cutaneous nerve**
  - Continuation of **musculocutaneous nerve**
  - Protect while raising flaps
  - Injury → main risk **painful neuroma**, more than sensory loss
- Anterior + posterior flaps raised

Superficial Dissection: Splitting the EDC Ask AI

EDC

- **Lateral muscles** exposed (anatomy covered previously)
- Possible **white line** visible
- Anterior to EDC → **extensor carpi radialis longus** + brevis
- Posterior to EDC → **extensor carpi ulnaris** + anconeus
- EDC middle → **intermuscular dissection** → EDC split
  - → dorsal part + volar part
- Beneath EDC → **joint capsule** + supinator muscle
  - Supinator → **posterior interosseous nerve** passes through it

Deep Dissection: Capsule and Supinator Ask AI

Supinator Opening Ask AI

PIN

- Open supinator with forearm in **pronation**
  - → PIN moves away → protected

Capsulotomy Ask AI

Capsule

- Capsule opened at **equator of the radial head** → middle of wound
  - Not anterior → **posterior interosseous nerve**
  - Not posterior → **lateral collateral ligament** attachment
- Capsule opened → joint exposed; **supinator** below
- Capsule at equator → **capitellum** seen

Supinator Release Ask AI

PIN

- Always in **pronation**, with scissors
- Cut nothing **not seen** with scissors
- → protects **posterior interosseous nerve**

Retraction Ask AI

PIN

Side	Retractor	Reason
Dorsal	Hohmann retractor	No problem dorsally
Anterior	Ordinary retractor, not Hohmann	PIN anterior; probably recurrent radial artery

- Anterior **Hohmann retractor** → leverage → traction injury to important structures

Exam Pearl

- Capsule opened at **equator of the radial head**
- Anterior → PIN; posterior → **LCL attachment**
- Supinator → opened in **pronation** with scissors
- Anterior retraction → ordinary retractor, never **Hohmann**

Proximal Extension: Lateral Column Ask AI

Indication Ask AI

Lat column

- EDC splitting widened **proximally**
  - → work on **lateral column of the distal humerus**

Landmarks and Interval Ask AI

Lat column

- Palpate lateral border of humerus → **lateral supracondylar ridge**
- Open along **lateral intermuscular septum**
  - Posterior: triceps muscle
  - Anterior: **mobile wad** muscles
- Mobile wad → **brachioradialis** + extensor carpi radialis longus

Origin Anatomy Ask AI

Lat column

- 1st from supracondylar ridge → **brachioradialis** → down to radial styloid
- 2nd from supracondylar ridge → **extensor carpi radialis longus**
- Below → lateral humeral epicondyle → **common extensor origin**
  - → extensor communis seen here

Superficial Layer Ask AI

Lat column

- Up along lateral supracondylar ridge
  - Triceps behind; **mobile wad** in front; both superficial
- **Triceps muscle** raised posteriorly
- Anteriorly → **brachioradialis** → ECRL beneath

Deep Layer and Subperiosteal Elevation Ask AI

Lat column

- Deep layer → **brachialis**
- All elevation **subperiosteal** → periosteal elevator
  - Soft tissue raised off lateral column of distal humerus
- Why subperiosteal
  - **Radial nerve** → between mobile wad (brachioradialis) and brachialis
  - Intermuscular dissection here → onto radial nerve
  - → **subperiosteal elevation of the brachialis** protects nerve
- Sequence: brachioradialis + ECRL origin raised → **brachialis** found
  - → subperiosteal elevation of brachialis → radial nerve protected
  - **Radial nerve** lies exactly at this spot
  - Any intermuscular dissection here → onto radial nerve

Flap Summary Ask AI

Lat column

- EDC split → raised from **lateral humeral epicondyle** → proximal
- **Brachioradialis** + ECRL raised
- **Brachialis** elevated subperiosteally medially → radial nerve protected
- **PIN branch** also seen here
- **Superficial radial nerve** → under brachioradialis
- Radial nerve: brachialis ↔ **mobile wad** (brachioradialis + ECRL)
  - Subperiosteal elevation from brachialis → **radial nerve** protected

Joint Access Ask AI

Lat column

- After elevation → **joint capsule** in front
- **Capsule detached from distal humerus** → good view of:
  - Radial head
  - Capitellum
  - Trochlea
- Dorsal view → **subperiosteal elevation of the triceps**
  - Posterior view → triceps elevated from posterior aspect
- **Common extensor origin** on other side → not detached
  - Only anterior part of common extensor origin released
- Capsule then raised

> **Clinical Pearl**
> • **Radial nerve** → between brachialis and mobile wad
> • Intermuscular dissection here → radial nerve injury
> • **Subperiosteal elevation** of brachialis → nerve protected

Exposure Summary and Nerve Protection Ask AI

Structures Protected Ask AI

Lat column

- Joint exposed
- **Brachialis** raised with extensor carpi radialis longus
- Nerves protected
  - **Radial nerve**
  - **Superficial radial nerve**
  - **Posterior interosseous nerve**

Supinator Handling Ask AI

PIN

- Distal in wound → **supinator** muscle
- Opening: snip with **forceps** → continue with scissors
- Cut nothing **not seen**
- Forearm in **pronation**
- No **Hohmann anteriorly**
  - Posterior on radial side → acceptable
  - Anterior traction → ordinary **retractor**

Distal Humerus Exposure Ask AI

Lat column

- Capsule detached from distal humerus → **anterior distal humerus** exposed
- Posterior work → **posterior distal humerus** also accessible

PIN Position with Forearm Rotation Ask AI

Pronation Ask AI

PIN

- **Posterior interosseous nerve** → anterior
  - Radial head, anteriorly displaced PIN, supinator seen
  - Radial head → nerve crossing, dorsally: approximately 4 cm
- Supinator dissection → cut nothing **not seen**

Neutral Ask AI

PIN

- PIN begins to appear in wound
  - In **pronation** → not seen; positioned anteriorly
  - No **Hohmann retractor** there → traction injury; ordinary retractor instead
- Seen with radial head, joint capsule, distal humerus, **supinator**
- Nerve approaches → safe distance at **proximal radius** decreases
  - Now 2.5 cm

Supination Ask AI

PIN

- Nerve appears more → right onto **field**
- Ruler: **crossing in the supinator** at 2 cm
  - Only 2 cm distal → immediate injury in supination

Rotation	PIN position	Radial head → nerve crossing
Pronation	Anterior → away from field	~4 cm
Neutral	Begins to appear in wound	2.5 cm
Supination	Right onto field	2 cm

High Yield

- PIN position **changes with forearm rotation**
- **Pronation** → PIN anterior → more distance
- Supination → 2 cm → highest risk
- More exposure → identify nerve; consider **lateral column approach**

Recap: EDC Splitting and Lateral Column Ask AI

Advantages of EDC Splitting Ask AI

EDC

- Reduced disadvantage and risk vs **Kaplan approach**
- Protects **lateral collateral ligament** vs dorsal (Kocher) approach
- Proximal extension → **mobile wad** + brachialis raised
  - → distal humerus + lateral column seen

Lateral Column Approach Ask AI

Lat column

- Whole lateral column → open proximally from **lateral humeral epicondyle**
  - 4–5 cm proximally along **lateral border of the distal humerus**
- Same arrangement: **extensor digitorum** → anterior: ECRL + brachioradialis
- Protect anterior flap → **lateral antebrachial cutaneous nerve**
  - Can be cut at this position
- Extensors + brachioradialis raised → **brachialis** beneath
- **Subperiosteal elevation of the brachialis** → protects:
  - Radial nerve
  - Superficial radial nerve
  - Posterior interosseous nerve
- **Radial nerve** → exactly at this spot
  - Blind dissection here → nerve cut
  - → dissection must be **subperiosteal**

Medial Continuation Ask AI

Lat column

- Medial + lateral borders of humerus identified
- **Hohmann retractor** safe here → subperiosteal plane protects nerve
- Humerus seen without **opening the capsule**
- Retraction continued
  - → until **medial side** reached
- Humerus, **coronoid fossa** + fat pad seen
- Anterior or posterior work in this area → no problem

Lateral Epicondyle Osteotomy Ask AI

Indications Ask AI

Osteotomy

- Further extension → **whole joint** exploration
  - **Fractures**
  - Contractures needing release
  - Any other required work
- → **lateral epicondyle osteotomy**

Technique Ask AI

Osteotomy

- Identify **lateral epicondyle**
- Up 4 cm → osteotomy fragment
  - 1 cm from lateral border dorsally
  - 1 cm from lateral border volarly
- **Chevron**, not transverse → as olecranon osteotomy; same cut direction
  - → screw fixation **stable**
- **Pre-drill** screw holes before cut
- Osteotomy → **lateral epicondyle** lifted
- Distal dissection: **extensor carpi ulnaris** ↔ anconeus → continued distally

Direction	Structures raised / interval
Anterior	Brachialis + brachioradialis + extensor carpi radialis longus
Dorsal	Triceps
Distal	ECU ↔ anconeus (modified Kocher approach) → osteotomy

- Whole lateral side open → **lateral humeral epicondyle** lifted with flap
- Whole joint explored; medial side reached by **valgus**

Ulnar Nerve Risk with Valgus Ask AI

Osteotomy

- Opposite, unseen side → **medial ligament** + ulnar nerve (more important)
  - Valgus → medial traction → injury to both
- Long-standing **valgus** → ulnar nerve palsy
  - Ulnar nerve not exposed medially → palsy still occurs
  - Excessive, long-standing valgus → **intermittent traction** → palsy

Pitfall

- Lateral exposure + valgus → **ulnar nerve** at risk medially
- No **excessive valgus**
- No prolonged valgus
- Otherwise → **ulnar nerve palsy**

Procedures Possible Ask AI

Osteotomy

- **Radial head** procedures
- Reconstruction of **capitellum** + lateral trochlea
- **Coronoid** visible
- Complex case → any needed work possible
- Key concern → **ulnar nerve**

Closure Ask AI

Osteotomy

- Pre-drilled fragment returned → **fixed**
- Anatomy repaired → wound closed

Lateral Approaches to the Elbow: Summary Ask AI

- Kaplan approach
- Modified Kocher approach
- Extensor digitorum communis splitting approach
- Lateral column approach
- Lateral epicondyle osteotomy approach
- Remaining → **Boyd approach**

Boyd Approach Ask AI

Technique Ask AI

Boyd

- Entry along **posterior border of the ulna**
- **Anconeus muscle** raised from ulna → radial head seen posteriorly

Indications Ask AI

Boyd

- **Posterior fracture**
- Posterior opening for ulna fracture → e.g. **Monteggia fracture**
- Posterior approach + anconeus raised → **Boyd approach**

Disadvantage Ask AI

Boyd

- Muscle elevation → entirely **subperiosteal**
- Ulnar periosteal stripping close to **proximal radius**
  - → irritation → new bone formation → **proximal radioulnar synostosis**

Pitfall

- Boyd: ulnar periosteal stripping near **proximal radius**
- → **proximal radioulnar synostosis**

Course Continuation Ask AI

- Next → **surgical anatomy** focus, given its surgical importance
- Then → **medial surgical approaches** to the elbow

Posterior Elbow Approach (Ulnar Nerve)

(Landmarks, Ulnar Nerve and Radial Nerve Protection)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Skin Incision and Superficial Dissection Ask AI

- Elbow course continued → **posterior approach to the elbow**
- Previous lecture → **lateral approach**; review optional
- **Single skin incision**
  - Total length ≈ 15 cm
  - ≈ 8 cm proximal + 7 cm distal to **olecranon tip**
- **Superficial dissection** → posterior skin flap raised as one block
  - No dissection between skin and subcutaneous tissue
  - → preserves flap **healing**
  - → protects **cutaneous nerves**

High Yield
- Incision ≈ 15 cm → 8 cm proximal, 7 cm distal to **olecranon tip**
- Posterior flap raised **as one block** → healing + cutaneous nerves

Anatomical Landmarks of the Posterior Elbow Ask AI

- Bony landmarks
  - **Olecranon tip**
  - **Lateral humeral epicondyle**
  - **Medial humeral epicondyle**
- Muscle → **triceps**
- Olecranon → divides region into lateral and medial sides
- Lateral to olecranon → **anconeus**
- Medial to olecranon → **flexor carpi ulnaris** (FCU)

Ulnar Nerve Exploration Ask AI

- Principle → first step = find **ulnar nerve**
  - Dissection in close relation to it
  - Exploration → prevents injury, postoperative **neuritis**, nerve palsy
- Surface anatomy → palpation posterior to **medial epicondyle**
  - → **cord-like** nerve felt
- Traced proximally → probably along **medial intermuscular septum**
  - Nerve rolls under finger → cord felt
- Extent of **exploration**
  - Proximal → 5–6 cm
  - Distal → into FCU, between its heads
- Distal landmark → **nerve supply to FCU**
- Distal exploration → risk of cutting **FCU nerve branch**
- Degree of release
  - Fully stripped nerve in image → **cadaveric specimen**, not patient
  - In practice → preserve surrounding **soft tissue and fat**

  - Preserve accompanying vessels, if present
  - Avoid nerve stripped like a **bare wire**

Must-Know
- First step of posterior approach → **ulnar nerve** exploration
- Exploration: 5–6 cm proximal → distal into FCU
- Distal landmark → nerve supply to FCU

Pitfall
- Distal exploration → **FCU branch** can be cut
- Complete stripping (cadaver image) → avoid in patients → **bare wire**

Radial Nerve and Proximal Extension Ask AI

- Lateral side → **radial nerve**
  - Posterior → **spiral groove**
  - → passes through **lateral intermuscular septum**
  - → emerges anteriorly between brachialis and **brachioradialis**
- **Proximal extension** → exposure widened upward
  - → **radial nerve** explored → protected
  - Not looked for → high risk of transection

Structures Defining the Posterior Approaches Ask AI

- More than one **posterior approach** → all relate to same structures
  - **Triceps**
  - **Olecranon**
  - **Anconeus**
  - **Flexor carpi ulnaris**
  - **Collateral ligaments**
- Approaches differ by **deep dissection** → what is opened, what is elevated

Nerves at Risk: Recap Ask AI

- Anterior + posterior views → **ulnar nerve** + radial nerve
  - Ulnar nerve → always looked for
- Proximal extension → e.g. **plate** for proximal fracture extension
  - → **radial nerve** explored → otherwise can be cut

Feature	Ulnar nerve	Radial nerve
Location	Posterior to medial epicondyle; cord-like	Spiral groove → lateral intermuscular septum → anterior, brachialis/brachioradialis
When explored	Always; first step	When extending proximally

Feature	Ulnar nerve	Radial nerve
Exploration extent	5–6 cm proximal; distal into FCU	—
Risk	Injury, neuritis, palsy; FCU branch cut	Transection during proximal extension

Pitfall

- Proximal extension (e.g. plate) → **radial nerve** must be explored
- Not explored → can be cut

Ulnar Nerve: In Situ vs Anterior Transposition Ask AI

- Question → handling of identified **ulnar nerve** in operated cases

Two Options Ask AI

Ulnar n.

- After exploration and protection
  - Option 1 → explore → leave **in situ**
  - Option 2 → explore → **anterior transposition**
- Which is better → to be decided before surgery

Meta-analysis (2018) Ask AI

Meta 2018

- Published in ***Hand Clinics*** → strong **hand surgery** journal
- Comparison → nerve left **in situ** vs anterior transposition
- **Ulnar neuropathy** → common in all distal humeral fractures
  - Overall incidence ≈ 19%, both groups combined
  - 19% of all cases → whether transposed or in situ
- **Postoperative ulnar neuropathy** ↑ after transposition
- Common belief → transposition prevents **postoperative ulnar neuropathy**
  - Meta-analysis → belief **not correct**
- **Anterior transposition** → higher neuropathy rate than in situ
- Included studies → cause of **neuritis** not stated

Possible Causes of Neuropathy After Transposition Ask AI

Meta 2018

- Not detailed in studies; possible causes
  - Original **fracture**
  - **Manipulation** during anterior shift → iatrogenic injury
  - Postoperative perineural **fibrosis** → compression
- Overall → most neuropathy cases after **anterior transposition**

Lecturer's Practice Ask AI

Ulnar n.

- ≈ 99% of cases → nerve left **in situ**
- **Ulnar neuritis** → very rare: ≈ 1 case per 100–200
  - Management → open + **release**, or improves with time
- In situ → check relation to **medial plate**
  - **Screws** or plate edge threatening nerve → transpose (very rare)
  - Otherwise → in situ in **fracture** cases; transposition rare

Clinical Pearl

- Fracture cases → ulnar nerve left **in situ** (≈ 99%)
- Neuritis ≈ 1 per 100–200 → release, or improves with time
- Plate screws/edge conflict → **transposition** (very rare)

Indications for Anterior Transposition Ask AI

- Non-fracture situations → nerve must be moved
  - **Elbow release** for contracture → regain range of motion
  - Any case that itself requires **anterior transposition**
- Transposition → **not prohibited**
- Debate limited to **fracture cases**

Randomized Controlled Trial (2021) and Recommendation Ask AI

- **Meta-analysis** → pools all literature studies on topic
  - Transposition studies + in situ studies
  - → **pooled analysis** → results compared

Trial Design Ask AI

RCT 2021

- Published in ***Journal of Orthopaedic Trauma***, 2021
- Comparison → **in situ placement** vs anterior transposition
  - Population → **distal humeral fractures** with plate fixation
- Design → **multicenter randomized controlled trial**
- **Level of evidence I**

Result Ask AI

RCT 2021

- **No significant difference** in outcome
  - Ulnar nerve mobilization + in situ placement
  - vs **anterior subcutaneous transposition**
  - After **bicolumnar plate fixation** of distal humerus fractures
- Both strategies → acceptable

Recommendation Ask AI

Ulnar n.

- Meta-analysis → indicates direction to go
- RCT → no difference
- → do the easier → leave nerve **in situ**
  - No difference → no need to transpose
  - Consistent with **meta-analysis** findings

Feature	Meta-analysis	RCT
Year	2018	2021
Journal	Hand Clinics	Journal of Orthopaedic Trauma
Design	Pooled analysis of literature	Multicenter RCT; Level I
Population	Distal humeral fractures	Distal humeral fractures; bicolumnar plate fixation
Finding	Neuropathy ≈ 19% overall; higher after transposition	No significant outcome difference

Exam Pearl

- Meta-analysis 2018 (*Hand Clinics*) → ulnar neuropathy ≈ 19%
- Transposition → **more** postoperative ulnar neuropathy
- RCT 2021 (*J Orthop Trauma*), Level I → **no difference**
- Conclusion → leave nerve in situ

Posterior Approaches: Defining Structures and Chronology Ask AI

- Approaches → defined by relation to **structures**
  - **Triceps muscle**
  - **Anconeus muscle**
  - **Olecranon**
  - **Flexor carpi ulnaris**
- Presented **chronologically**
  - 1931 → **Campbell**, textbook author

Triceps Splitting Approach

(From Campbell to the Triceps-Splitting Variants)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Posterior Approaches to the Elbow Ask AI

- Classified by relation to **local structures**:
  - Triceps muscle
  - Anconeus muscle
  - Olecranon
  - Flexor carpi ulnaris (FCU)
- Chronologically first → **Campbell**, 1931
  - Author of *Campbell's Operative Orthopaedics*

Campbell's Original Approach (1931) Ask AI

Technique Ask AI

Campbell

- Aim: open elbow **from behind** → reach front of joint
- Posterior midline opening → triceps attached to olecranon
- **Triceps tendon stripped** off olecranon → released completely

Exposure Ask AI

Campbell

- **Distal articular surfaces** well seen
- Lateral collateral ligament released → articular surfaces seen more fully
- Joint can be **dislocated** → everything readily accessible

Objection Ask AI

Campbell

- Tendon stripped **completely** off olecranon
- Tendon sutured back → problems:
  - **Extension** problems
  - Problems from the approach technique itself
- → Later surgeons introduced **modifications**
- Stripping performed → **tendon repair** mandatory
  - Secure fixation, e.g. wires → good healing
  - Otherwise → same problems recur
- Later search → method avoiding complete **tendon stripping**

High Yield

- Campbell 1931 → triceps tendon stripped completely off olecranon
- Sutured back → extension problems
- Stripping done → secure tendon repair, e.g. wires

Extensile Options Ask AI

Campbell

- Origin of the ideas behind the **Triceps Splitting Approach**
- Can be made **extensile**:
  - Anconeus raised → lateral compartment entered from posterior → **radial head** exposed
  - FCU raised medially → **anteromedial facet** reached

Current Status Ask AI

Campbell

- Complete stripping preferred? → **No**; practically nobody does it
- Not favoured — even by **Campbell himself**

Campbell's Triceps Tendon Reflection (Tongue) Ask AI

Technique Ask AI

Tongue

- Later Campbell description → **triceps tendon reflection** (tongue)
  - Portion of triceps tendon + part of muscle, across **full dimension**
  - Portion **reflected** like a tongue → exploration
  - Muscle portion left attached to tendon

Exposure Ask AI

Tongue

- **Lateral column** well seen
- Wider exploration → lateral ligaments released
- Released ligaments **re-inserted** → stability
  - Transosseously, or
  - With anchors
- Further anterior exploration → **capsular release** possible
- If forced → **collateral ligament** release, when that side needed

Pitfall

- Released collateral ligaments → must be re-inserted
- Transosseous or anchors → stability restored

Rationale and Indications Ask AI

Tongue

- Previous method → problems → chosen to **reduce complications**
- Especially useful in **contracture** cases
- Fibrosis + contracture → muscle **shortening**
- → **Triceps tendon lengthening** → flexion range increased
  - Length gained throughout the muscle
- Used by lecturer in the past; still used today → open, fix
- Results **good**; no problems seen in practice
- **No longer used much** nowadays; mentioned for awareness

Triceps Splitting Approach Ask AI

Rationale Ask AI

Splitting

- Criticisms of stripping/reflecting the whole muscle:
  - **Fibrosis**
  - Repair problems
  - Rupture
- Proposal → **splitting approach**
  - Respects the muscle; tissues kept in good condition

Technique Ask AI

Splitting

- Usual **posterior incision**, about 15 cm
- **Ulnar nerve** exposed
- Tendon (substantial **fibrous band**) raised from distal triceps muscle
- **Fibrous band only** raised
  - Fleshy part left across full triceps width
- Flap left **inserted distally** onto olecranon
- Beneath → two triceps heads: **deep head** and lateral head
  - Substantial fibrous cord/band between them

Dissection Between the Heads Ask AI

Splitting

- Fibrous band split **in the middle**
- Lateral part peeled **off olecranon**
- Medial part retracted
- → **Distal humerus** well exposed

Extension and Closure Ask AI

Splitting

- Anterior access → lateral or medial **collateral ligament** released → re-inserted
- Distal extension → **anconeus** raised (likely from the ulna)
- Procedure complete → muscle **closed** back as it was
- Re-insertion needed → **transosseous sutures**

Triceps Lengthening Alternative Ask AI

Splitting

- No re-insertion + lengthening wanted → alternative to **V-Y (tongue)** shape
  - Indication: **elbow contracture** opened posteriorly → contracture released → triceps needs lengthening
- Lengthening technique:
  - One part attached medially, the other laterally
  - Sutured to **raised flap**, not bone → length gained
- Practically **laborious**
  - Raising + suturing → prolonged operation
  - Unless triceps tendon lengthened by another method

Clinical Pearl

- Contracture → fibrosis → triceps shortening
- Triceps lengthening → flexion range increased
- Options → tongue (V-Y) reflection or splitting flap sutured to flap

Medial Splitting Modification Ask AI

- Splitting modification within the posterior approach
- Previous modification → **tongue shape**
- Aponeurosis + muscle split from **medial side**
  - Alongside the ulnar nerve
  - One-third of muscle left; two-thirds raised laterally

Blood-Supply-Respecting Modification Ask AI

Rationale Ask AI

BS-sparing

- Posterior blood supply of triceps → **poor**
  - Depends on **anastomosis from anterior**
- → Triceps opened to respect the blood supply

Exam Pearl

- Triceps posterior blood supply → poor
- Depends on anterior anastomosis → opening must respect it

Technique Ask AI

BS-sparing

- Posterior opening; **ulnar nerve** exposed
- Ulnar nerve side → medial; opposite side → lateral
- Landmarks: triceps muscle, olecranon, **proximal ulna**
- Fibrous band flap raised **from lateral**
  - Not from medial, as in the previous approach
- Lateral part of triceps raised together with it
- After flap raised:
  - Ulnar nerve exposed
  - **Two muscles** separated from each other
  - Cord between them **released** → good visualisation
  - Muscle raised → **humerus** exposed
- Distal portion left on **olecranon**
- If needed → **anconeus** raised

Outcome Ask AI

BS-sparing

- Advantage → **respects blood supply**; reference paper available
- Authors' report → good results, no problems

Comparison of Posterior Approaches Ask AI

Approach	Triceps handling	Key point
Campbell original (1931)	Tendon stripped completely off olecranon	Extension problems; secure repair, e.g. wires
Tongue reflection	Tendon + muscle portion reflected, full dimension	Triceps lengthening in contracture; no longer used much
Triceps splitting	Fibrous band split in middle; flap inserted distally on olecranon	Respects muscle; avoids fibrosis, repair problems, rupture
Medial splitting modification	Split from medial side alongside ulnar nerve	One-third left; two-thirds raised laterally
Blood-supply-respecting	Flap raised from lateral	Respects anterior anastomotic blood supply

Triceps Sparing Approach

(Exposure of the Distal Humerus with the Triceps Intact)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 116) “between anconeus and flexor carpi ulnaris” → “between anconeus and extensor carpi ulnaris”

Alonso-Llames (Bilateral Tricipital) Approach Ask AI

Concept Ask AI

Alonso

- Evolution: **Campbell** approach + modification → Alonso-Llames
- Principle: no dealing with the **triceps** at all
  - Triceps left entirely intact
  - Whole **insertion on the ulna** preserved
- Steps
  - Medial side → **ulnar nerve** exploration → elevate muscle
  - Lateral side → elevate muscle
  - → **distal humerus** well visualised

Indications Ask AI

Alonso

- Original use: **extra-articular fractures** → supracondylar distal humerus, children
  - Extra-articular → **K-wires** + reduction, no problems
- Adults → works well in most cases
  - Even **intra-articular fractures**
- Condition: no **severe comminution** (e.g. not 3–4 fragments)
  - **Supracondylar–intercondylar fracture**, 2 distal fragments → works well
  - More than 2 fragments → possible if anterior view not needed
  - Anterior part → not visible through this approach
- Joint exploration → limited
- Visualisation → modest only

Reduction Technique and Rotation Control Ask AI

Tension Problem Ask AI

Alonso

- Elbow **flexion** + muscle elevated both sides + triceps attached → triceps tension
- Lifting the muscle → requires elbow **extension**
  - Extension → risk of **fracture displacement**
- Supracondylar–intercondylar fracture: medial + lateral elevation → fracture lies in front

Reduction Steps Ask AI

Alonso

- Goal: 2–3 distal fragments → convert to **one fragment**
  - If needed → extend the elbow
  - **Anconeus** → elevate a little laterally
  - **Flexor carpi ulnaris** → elevate a little medially
  - Explore **articular surface** → reduction
- Important point to watch → especially with this fracture pattern

Rotation Check Ask AI

Alonso

- Distal fragment → may be **malrotated** unnoticed, even while looking fine
- Rotation check → very important in every case treated
- Signs of no rotation
  - **Olecranon fossa** → lined up well
  - **Anterior cortex** of humerus, medial + lateral sides
  - Both → lined up with each other
- K-wires inserted → transverse component fixed → K-wires fixed to proximal fragment → image
  - **X-ray**: distal fragment in flexion or extension?
  - Any rotation?
- Most important intraoperative point → distal fragment has **no rotation**

High Yield

- Distal fragment can be malrotated while looking fine
- No rotation → **olecranon fossa** lined up
- No rotation → **anterior cortex** medial + lateral lined up
- Post-fixation X-ray → flexion/extension of distal fragment + rotation

Single K-Wire Trick Ask AI

Alonso

- Two articular fragments reduced → **one preliminary K-wire** only, not a screw
- → reduce **metaphyseal fracture** to proximal fragment, lateral side
- → confirm reduction **anteriorly and posteriorly**
- → no **rotation** in either component
- Residual rotation → corrected by the reduction itself
  - **Reduction clamp**
  - **Hohmann**
  - **Bone forceps**
- Most cases → good reduction, no problems
- Extra **third fragment** → also manageable
- Manipulation: flexion → **tendon** tension → field closes
- → **preliminary reduction** → extend → field widens, tendon relaxes

Clinical Pearl

- Articular fragments reduced → **one preliminary K-wire**, not a screw
- Then metaphyseal component reduced to proximal fragment
- Anterior + posterior reduction confirmed → no rotation either component
- Flexion closes the field → preliminary reduction, then extend

Triceps-Sparing Approach: Advantages and Evidence Ask AI

Extensibility Ask AI

Alonso

- Option retained → cannot work well → convert
- Extra-articular bone strip from **proximal ulna** possible
  - Reattachment → bone-to-bone → better **repair**
- Conversion / extension options
  - **Anconeus** → elevate laterally
  - **Flexor carpi ulnaris** → elevate medially
- Lecturer's view: fracture → start with this approach
  - More needed → field already open
- **Olecranon osteotomy** → lecturer dislikes, always avoids
  - Seen on X-ray or report → not good in his view
  - Still an available option

Nomenclature Ask AI

Alonso

- **Triceps-sparing approach**
- **Bilateral tricipital approach** → triceps elevated from medial + lateral sides
- Literature review questions
  - Worthwhile, or just copying what we read?
  - Easier?
  - Better results?
  - Fewer **complications**?
  - Answers → from users producing the **evidence**

Evidence Ask AI

Alonso

- Triceps-sparing users report:
  - **Does not compromise the ligamentous joint stability** → medial + lateral stability intact
  - Nor **the stabilizing effect of the anconeus muscle** → unless elevated
    - Elevation optional → field often good without it
    - Anconeus → very important for the **elbow**
- Nothing irreversible → more can be done if needed
- Available in **children** and adults

Must-Know

- Triceps-sparing = **bilateral tricipital** = Alonso-Llames approach
- Ligamentous joint stability → not compromised
- Anconeus stabilizing effect → preserved unless elevated
- Nothing irreversible → convertible / extensible

Limitations and Preoperative Planning Ask AI

Limitations Ask AI

Alonso

- **Comminution** of distal humerus, especially anterior → not visible
  - Flexion → tendon closes field
  - Extension → still no anterior view
- Anterior comminution, e.g. **capitellum** displaced anteriorly → not retrievable
  - → another approach needed
- **Coronal fracture** of the trochlea → same limitation
- → **preoperative planning** before choosing the approach

Pitfall

- Anterior comminution → not seen in flexion or extension
- Anteriorly displaced **capitellum** → not retrievable
- Coronal trochlear fracture → same
- Plan the approach preoperatively

Literature Appraisal: Triceps Splitting vs Sparing Ask AI

The Study Ask AI

Item	Detail
Title	"Comparison of outcomes after triceps splitting versus sparing"
Year	2014
Journal	Injury (name uncertain)
Design	Retrospective study
Groups	Triceps splitting vs triceps sparing
Outcome	Later elbow function

- Published 2014 → retrospective splitting vs sparing review
- Conclusion: **extra-articular distal humerus fractures** → triceps sparing **can result in better elbow range of motion and triceps strength than the triceps splitting approach**

Critical Appraisal Ask AI

- Common belief → but **study design** matters → know its limitations
  - Adopt its conclusion or not?
  - Consider **level of evidence**
  - Own critical view of the paper
- Limitation 1: **retrospective review** → cannot really settle the question
- Limitation 2: **multiple variables** → yet concludes sparing better than splitting
  - Which triceps splitting was done?
  - **Fracture complexity**
  - Outcomes assessed → affected by other variables too
    - **Range of motion**
    - **Triceps extension strength**
    - **DASH score**
  - Confounders
    - Fracture complexity
    - **Muscle injury** in the elbow
    - **Bone healing**
    - **Fixation** method
- → attributing outcome difference to approach alone → not valid
- Study not dismissed → judge reliability from authors' **methodology**

Exam Pearl

- 2014 splitting vs sparing study → **retrospective**
- Multiple variables → difference not attributable to approach alone
- Confounders: fracture complexity, muscle injury, bone healing, fixation
- Reliability judged from the **methodology**

Bryan-Morrey Approach Ask AI

Technique Ask AI

Bryan

- **Triceps** + tendon → taken from **ulnar (medial) side** → elevated laterally
  - Whole triceps elevated from medial side
  - → continues with **periosteum** of proximal ulna
  - → triceps detached from **olecranon**
- No olecranon bone piece
  - Periosteum of proximal ulna → in continuity with **anconeus** laterally

Exposure and Rationale Ask AI

Bryan

- Joint exploration → good
  - Optional **dislocation**
  - Medial + lateral **release** → full access
- Drawback → triceps still detached from the **ulna**
- Idea → better exposure
- Rationale → muscle left in continuity with proximal ulna periosteum → reattachment
  - → results better than the previous approach

Anconeus–Lateral Triceps Flap Approach Ask AI

Concept Ask AI

Anc-LT

- Not a **splitting approach**
- Not like **Bryan-Morrey** → removes triceps tendon from proximal ulna
  - In continuity with proximal ulna periosteum + anconeus

Anatomy Ask AI

Anc-LT

- Structures in the images
  - **Medial triceps** → medial
  - **Lateral triceps** → lateral
  - **Anconeus** → lateral
  - **Flexor carpi ulnaris** → medial
  - **Ulnar nerve**

Technique Ask AI

Anc-LT

- **Triceps tendon** → interval between central part and the two heads
- Enter lateral to central part → separate the two heads
  - Lateral muscle → to be elevated
- Lateral muscle → separated from medial part
- Proper triceps → left with medial part
- Open posteriorly → interval between two heads → lateral side → elevate **lateral triceps**
- Incision + heads separated → move distally on the **forearm**
  - Landmarks: ulna, olecranon, **anconeus muscle**
- Interval: anconeus ↔ **extensor carpi ulnaris** (anterior, lateral)
  - = **modified Kocher approach**
- Distal part of modified Kocher opened
  - **Anconeus flap** + lateral triceps → detached from proximal ulna as one flap
  - Anconeus → detached from proximal ulna
- Kocher interval (anconeus ↔ extensor carpi ulnaris) → anconeus elevated **subperiosteally** from ulna *[C1 · transcript: "between anconeus and flexor carpi ulnaris"]*
- Past the elbow → **capsule** opened → anconeus + lateral triceps elevated
- → elbow exposed, especially with **dislocation**

Exposure and Applications Ask AI

Anc-LT

- Exposure
  - **Distal humerus**, medial + lateral sides
  - **Radial head**
  - **Olecranon**
- Uses
  - **Arthroplasty**
  - **Release**
  - **Fracture** fixation
- Idea: preserve triceps + exposure of **articular surfaces** → fixation
  - Fixation, arthroplasty, or anything else
- **Preoperative planning** before choosing the approach → how will you open it?
  - Not deciding in theatre from the X-rays
  - Night before / at decision → exact plan known
- Fracture fixation through this approach
  - Distal → **forearm**; proximal → arm
  - Anconeus opened distally + elevated → **distal humerus** + fracture visible
  - Medial side → elevate medial part → work through it
  - Fragments → manipulable while open
  - **Plate** → medially and/or laterally if needed
- Lecturer: not used for fractures → prefers **Alonso-Llames approach** (bilateral tricipital)
  - Users report → good, no problems
- **Triceps sparing** → no muscle taken
  - Central part of tendon → left attached to ulna

Case Ask AI

Anc-LT

- **Metaphyseal comminution** → anconeus–lateral flap approach → good reduction
- Paper on this approach → good, no problems

Choosing an Approach Ask AI

- Complex / difficult → not necessarily to be done
- Choose the **easy** option
  - Gives a result
  - Few problems
  - Suits the case
  - Based on aims + **preoperative plan**
- Difficult option to look capable → wrong; goal = result for the **patient**

Approach	Triceps	Exposure	Notes
Alonso-Llames (triceps-sparing / bilateral tricipital)	Intact; elevated medial + lateral; ulnar insertion preserved	Distal humerus; anterior not seen; joint exploration limited	Extra-articular; intra-articular with `2` distal fragments

Approach	Triceps	Exposure	Notes
Bryan-Morrey	Triceps + tendon detached from olecranon, medial → lateral, with proximal ulna periosteum + anconeus	Good; dislocation + release possible	Triceps detached from ulna
Anconeus–lateral triceps flap	Central tendon attached to ulna; lateral triceps + anconeus as one flap	Distal humerus, radial head, olecranon	Arthroplasty, release, fracture
Anconeus approach	Not elevated unless extended	Lateral posterior distal humerus + radial head; medial not seen	Posterior radial head; capitellum

Anconeus Approach Ask AI

Concept Ask AI

Anconeus

- **"Extensile posterior approach to the elbow joint: the anconeus approach"**
- Anconeus elevated → triceps not elevated with it, unless situation progresses
- **Extensile** → opens elbow posteriorly

Anatomy and Exposure Ask AI

Anconeus

- **Topographic anatomy**: distal humerus, proximal part, lateral + medial sides, anconeus
- **Interval** → anconeus ↔ extensor carpi ulnaris
- Anconeus elevated → **lateral posterior component of the distal humerus** + radial head

Indications Ask AI

Anconeus

- **Radial head fracture** → fixable through this approach
  - Especially **posterior** radial head fracture → difficult via modified Kocher
- **Capitellum** fracture of lateral distal humerus
- Medial side → not visible
  - → extend proximally → **anconeus–lateral flap**
- Also cited for radial head and lateral condyle fractures (wording uncertain)
- Choose the **easy** approach → known + confident of good result

Anconeus–Triceps Hemi-Flap Approach Ask AI

- Name: **anconeus hemi-flap approach** (name uncertain)

Technique Ask AI

Hemi-flap

- Posterior opening over **proximal ulna**
- Anconeus → in interval with **extensor carpi ulnaris**
- Interval opened → anconeus elevated from ulna → flipped → **radial head** visible
- **Distal humerus** + capitellum visible
  - Anteriorly + posteriorly
  - Further anterior if wanted
- Published: **"Anconeus–triceps hemi-flap approach: a safe and effective technique"** → exposure; fine
- Do the **easy** option you know well

Olecranon Osteotomy Approach

(The Chevron Osteotomy and Its Alternatives)

Prof Dr Galal Mansour

Watch Original Lecture

Lecture transcript processed with AI assistance.

Contents

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 129) “intra-articular distal radius fractures” → “intra-articular distal humerus fractures”

Olecranon Osteotomy — Technique Ask AI

Position of the Approach Ask AI

Olecranon

- **Olecranon osteotomy** → established approach
- Lecturer personally dislikes it
- Sometimes forced → needed to visualise the joint, fracture-dependent

Osteotomy Site Ask AI

Olecranon

- Key trick → **non-articular part** of olecranon
  - Also called the **bare area**
  - Lies between coronoid part of proximal ulna → **olecranon process**
- Osteotomy passes through this non-articular part
- Shape → **chevron** (V-shaped) osteotomy
  - Follows the shape of the articular border

Steps Ask AI

Olecranon

- Open → **ulnar nerve** exploration
- Open the joint
  - **Anconeus** → elevated laterally
  - **Flexor carpi ulnaris** → elevated medially
- **Forceps** passed into the joint → rests at the bare area
  - Forceps level = osteotomy level
- Cut started with **saw** → completed with osteotome
- Fragment reflected proximally → whole **distal humerus** exposed
  - Includes posterior part
  - **Flexion** → further view
  - Exposure → no problem
- **Preliminary drilling** (pre-drilling) optional
  - At fixation → implant enters the pre-made drill hole

High Yield

- Osteotomy through the **bare area** → non-articular olecranon
- Bare area → between coronoid part and olecranon process
- **Chevron** (V-shaped) cut → follows articular border
- Forceps in joint → marks osteotomy level
- Saw starts → **osteotome** completes

Case Examples and Fixation Pitfalls Ask AI

- Operated case → fixation shown
- Case without need for further intervention
  - In reality → **split** part present → must be addressed
- Case done well → no problem
- Next case → mistake: **short** fixation → later problems

  - **Construct failure** → complication
    - → **second operation** needed
  - Implant must be **long** → reaches far enough distally → well anchored
- After osteotomy → exposure as shown

Pitfall
- **Short** implant after olecranon osteotomy → construct failure
- Failure → **second operation** for the complication
- Implant must be long → distal anchorage

Chevron versus Transverse Osteotomy Ask AI

- One author → different cut instead of **chevron**
  - Argument → if no problem results, why not
  - Some cases → no problems at all
- Literature reviewed → how others think and choose
- **Comparative study**, retrospective
  - **Chevron osteotomy** vs **transverse osteotomy**
  - Conclusion → **no difference**

Step-Cut Olecranon Osteotomy Ask AI

Concept Ask AI

Step-cut

- **Anatomical study** → describes step-cut olecranon osteotomy
- **Extra-articular** osteotomy instead of intra-articular
  - **Triceps tendon** elevated with the fragment
- Same idea as triceps detachment pulling a bone piece → acceptable
- **Large fragment**, cut in a step
  - On reduction → **stable**, strong
- Purpose-designed **device** → guides the osteotomy

Device Technique Ask AI

Step-cut

- Device applied → fixed with **K-wires**
- Device imaged → confirms **articular-safe**
- Osteotomy started with saw → completed with **osteotome**
- Neat-looking result → not a reason to be impressed

Lecturer's Criticism Ask AI

Step-cut

- Remaining olecranon left → if **detached** during work → no bail-out
- Purely **anatomical (cadaveric) study**
  - No clinical use by anyone
  - Shown to say → **do not perform** this osteotomy
- Configuration → added **risk of fracture**
  - Multiple fragments → may become 3 fragments
- Good joint view, but **risky**
  - Surgeon must anticipate fragment breakage

	Chevron osteotomy	Step-cut osteotomy
Level	Intra-articular, through bare area	Extra-articular
Fragment	Fits together	Large fragment, step-cut; triceps tendon elevated with it
Reduction	—	Stable, strong
Guidance	Forceps at bare area	Dedicated device, K-wires, imaging
Evidence	Clinical use; compared with transverse	Anatomical (cadaveric) only
Fracture risk	Fragments fit together	Added risk → may become 3 fragments

Pitfall

- **Step-cut osteotomy** → cadaveric only, no clinical use
- Remaining olecranon may detach → no bail-out
- Added fracture risk → may become 3 fragments
- Lecturer's message → do not perform it

Lateral Epicondyle Osteotomy Ask AI

Literature and Documentation Ask AI

Lat epicond

- **Lateral epicondyle osteotomy approach** vs olecranon osteotomy → does it have a role?
- Most surgeons → do not **record** work, outcomes, results
  - Experience → **not documented**
  - Advice → **document results** so they are useful
- Approach covered in the **lateral approaches** lecture → refer back
  - Images compared vs olecranon osteotomy
- **Articular surface** visible → anatomical study, same idea

Technique Ask AI

Lat epicond

- Open laterally → explore:
  - **Lateral border** of distal humerus
  - **Lateral humeral epicondyle**
  - **Extensor origin**
- Elevate soft tissue → clear lateral border
  - **Triceps** → posteriorly
  - **Brachialis** → anteriorly
- Osteotomy → reflected with the **common extensor origin**

Exposure Ask AI

Lat epicond

- Very good lateral exposure of the elbow
  - **Radial head**
  - **Distal humerus**
  - **Capitellum**
- **Varus** stress → clearer view
  - Caution → varus stress → **medial-side** problem
- Articular surface seen anteriorly, posteriorly, distally, laterally (radial head)

Lateral Epicondyle versus Olecranon Osteotomy — Exposure Study Ask AI

- Comparison paper (images from the paper)
  - **Olecranon osteotomy** vs **lateral epicondyle osteotomy**
- **Anatomical study** → measures visible articular surface per approach

Findings Ask AI

- **Lateral osteotomy (group L)**
  - Large part of **anterior** distal humerus seen → lateral viewpoint
  - Distal part → **distal lateral** only
  - **Medial** side not seen → unless excessive varus
    - Excessive varus → **medial-side** problem
- **Olecranon osteotomy**
  - More of the **posterior** part seen
  - Less anterior → opening from behind
  - Anterior articular surface of distal humerus seen → small
    - More seen with **flexion** and **distraction**
- Author's conclusion → each shows a certain part better
  - Neither preferred over the other
  - Exception → **whole distal humerus** needed, medially and laterally
    - → **olecranon osteotomy** (lecturer dislikes it)

	Lateral epicondyle osteotomy (group L)	Olecranon osteotomy
Better seen	Anterior distal humerus	Posterior part
Distal surface	Distal lateral only	Medial and lateral
Limitation	Medial side not seen	Small anterior view
Extra manoeuvre	Excessive varus → medial-side problem	Flexion + distraction → more seen
Indication	—	Whole distal humerus needed

Triceps Tongue versus Olecranon Osteotomy — Cadaveric Study Ask AI

- **Articular exposure of the distal humerus** → triceps tongue vs olecranon osteotomy
  - Anatomical, not clinical → **cadavers**
  - Published 2023
  - Conclusion → **no difference**
    - Same exposure with tongue as with osteotomy
- Studies presented for further reading
- Comparison also made → with vs without a **release** (side likely lateral)

Triceps Tongue versus Olecranon Osteotomy — Matched Cohort Study Ask AI

Triceps Tongue Recap Ask AI

Tongue

- Open posteriorly → **distally based tongue** of the tendon
  - Reflect → inspect joint

Study Details Ask AI

Tongue

- **Triceps tongue** vs olecranon osteotomy
  - Year → 2022
  - Centre → **Cleveland**
  - Country → USA
  - Design → **matched cohort study** (important term)

Retrospective versus Matched Design Ask AI

- **Retrospective study** → conclusion unreliable
  - Cause → **multiple variables** interfere with result
    - **Fracture complexity**
    - **Associated injuries**
    - **Mechanism of injury**
    - **Bone healing**
    - **Method of fixation**
- Example → retrospective **splitting** vs **bilateral tricipital** paper → conclusion unreliable
  - That paper → **limitation**
- **Matched** → author unifies (eliminates) these variables
  - Fixation → **plate** vs plate; **double plating** vs double plating
  - **Comminution** → extra-articular 2-fragment vs 2-fragment, and so on
- All cases alike in result-affecting variables
  - → no **significant difference** between groups
  - **Only variable** → approach: triceps tongue vs olecranon osteotomy

Exam Pearl

- **Retrospective study** → multiple variables → unreliable conclusion
- Variables → fracture complexity, associated injuries, mechanism, bone healing, fixation
- **Matched cohort** → variables unified between groups
- Only remaining variable → the approach

Methods Ask AI

Tongue

- **Retrospective** comparison → cases 2007 to 2019, different groups
- Not **multicentre** → multicentre = three or more centres
  - **Two separate institutions** → still respectable
- Triceps tongue performed up to 2019 → **recent** data

Results Ask AI

Tongue

- **Functional outcomes** → no difference
- Tongue approach showed:
  - **Increased range of motion** vs olecranon osteotomy
  - **Decreased intraoperative blood loss**
  - Lower rate of postoperative **ulnar nerve neuritis**

Author's Conclusion Ask AI

Tongue

- **Triceps tongue approach** → safe, reliable first-line approach for intra-articular distal humerus fractures *[C1 · transcript: "intra-articular distal radius fractures"]*
  - **Adequate visualisation** of the articular surface
  - Eliminates osteotomy-related complications
    - **Delayed union or non-union**
    - **Hardware failure or irritation**
  - Easy conversion to **total elbow arthroplasty**
- Published, 2 institutions, **Cleveland** → credible → recommended

	Triceps tongue	Olecranon osteotomy
Functional outcomes	No difference	No difference
Range of motion	Increased	Lower
Intraoperative blood loss	Decreased	Higher
Postoperative ulnar nerve neuritis	Lower rate	Higher rate
Osteotomy complications	Eliminated (delayed union/non-union, hardware failure/irritation)	Present
Conversion to TEA	Easy	—

Must-Know

- **Triceps tongue** → safe, reliable first-line for intra-articular distal humerus fractures
- Increased ROM, decreased blood loss, lower ulnar nerve neuritis
- No osteotomy complications → delayed union/non-union, hardware failure
- Easy conversion to **total elbow arthroplasty**

Choosing an Approach — Bottom Line Ask AI

- Do what you know how to do **well**
  - Easy, with good results
- Better option desired → **learn it** first → then perform it
- Difficult option for its own sake → not a mark of skill
  - Difficult option → **many problems**, intraoperative and postoperative

Olecranon Osteotomy versus Paratricipital Approach — Systematic Review Ask AI

- **Systematic review** → olecranon osteotomy vs **paratricipital approach**
  - Paratricipital → triceps elevated from both sides
- Conclusion → **choice of surgical approach does not influence outcomes**
  - More articular surface seen with one approach → outcome unchanged
- Author's message → do what you know **well**

  - **Easier** familiar approach → good result
- Case genuinely requiring an unfamiliar approach
  - → **learn it**, or
  - → **refer** to someone who can

Clinical Pearl

- Approach choice → no influence on outcomes (systematic review)
- Do the **easier** approach you know well
- Difficult option → many intraoperative and postoperative problems
- Unfamiliar but required approach → learn it or **refer**